Supplementary materials

Prediction models for highly scalable technology assisted differential diagnostics of ASD

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Contents


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## [1] "R version 4.5.2 (2025-10-31)"

## [1] "tidyverse version 2.0.0"
## [1] "knitr version 1.51"
## [1] "ggpubr version 0.6.2"
## [1] "ggrain version 0.1.0"
## [1] "Hmisc version 5.2.4"
## [1] "rstatix version 0.7.3"
## [1] "emmeans version 2.0.1"
## [1] "flextable version 0.9.10"
## [1] "officer version 0.7.2"
## [1] "kableExtra version 1.4.0"
## [1] "english version 1.2.6"
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S1 Inclusion and exclusion criteria

Inclusion criteria applying to all participants:

- no neurological diagnoses
- at least 18 and at most 60 years of age
- normal or corrected-to-normal vision
- legally binding, written informed consent from the participant for study participation

We only included clinical participants with either a BPD or an ASD diagnosis but excluded anyone who was given both diagnoses.

Additional inclusion criteria for ASD participants:

- valid F84 diagnosis according to ICD-10
- normal speaking abilities

Additional inclusion criteria for BPD participants:

- valid F60.31 diagnosis according to ICD-10
- no F84 diagnosis according to ICD-10
- IQ estimate above 70

Additional inclusion criteria for comparison participants:

- IQ estimate above 70
- no psychiatric diagnoses
- no current psychotropic medication

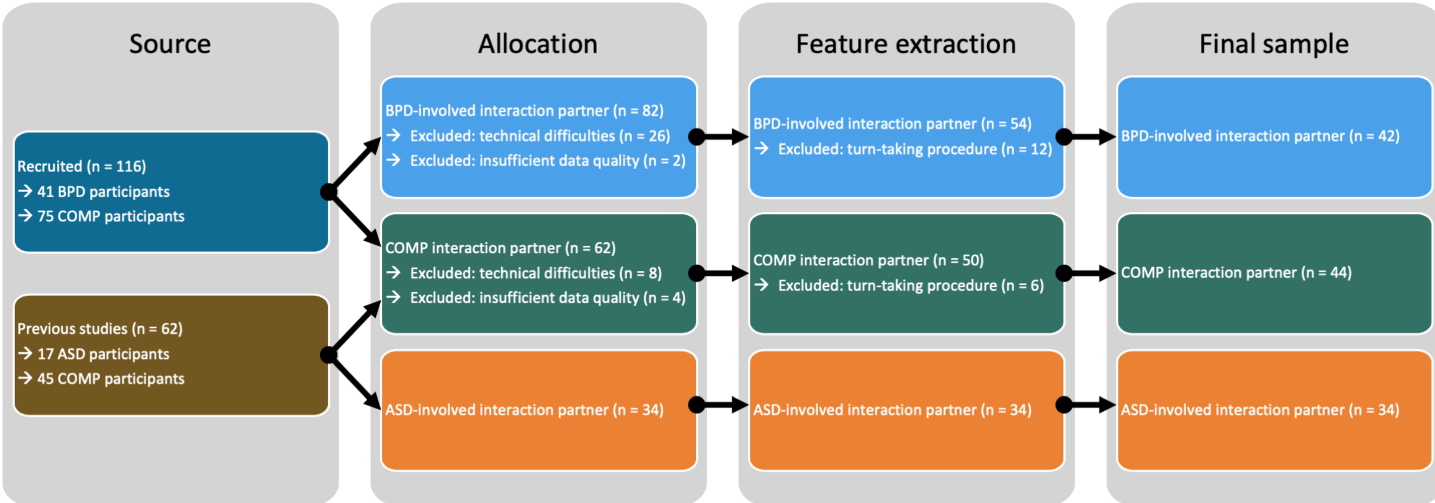
Furthermore, we applied the following exclusion criteria:

- underage people or people older than 60 years
- inability to speak
- current or previous neurological disorder
- acute suicidality, self-harming or aggressive behaviour
- lack of legally binding informed consent

Testing was discontinued if participants withdrew their consent, exclusion criteria applied or in the case of technical difficulties.

S1.1 Consort chart

Consort chart detailing the source of the data, exclusion of data and sample sizes for each interaction context. We reused data of ASD and comparison participants without psychiatric diagnoses (COMP) from two previous studies[1,2]. The data of all participants that were included in the analyses of both of these studies was included. Additionally, we recruited BPD and COMP participants. In a first step, we allocated participants to BPD-involved or COMP dyads based on their availability. Data of dyads where audio or video data of at least one interaction partner was missing was excluded. Due to a technical failure of our audio recording setup which had to be replaced, these numbers are substantial for the BPD-involved interaction contexts. Furthermore, we visually and aurally inspected the data and excluded data from dyads with insufficient data quality, for example, if a microphone picked up both interaction partners. After using the uhm-o-meter[3,4] to distinguish between each interaction partner speaking and being silent, we visually inspected the results and excluded data from all dyads where speaking turns were not correctly identified.



S2 Procedures

The hobbies prompt was chosen to capture monologuing by ASD adults when talking about their interests[5], while the mealplanning conversation required increased turn-taking and collaboration from both participants[6]. The order of the conversations was counterbalanced across participants and the experimenter left the room during the conversations. After both conversations were finished, they filled out a custom rapport questionnaire.

All conversations took place in stable artificial light to control for any biases caused by lighting changes[7]. Since some of the data was collected during the COVID-19 pandemic, we modified the setup to maximise the safety of our participants, for instance by placing a plexiglass between them. Furthermore, participants filled out several questionnaires to capture subjective clinical self-ratings, completed two intelligence assessments which were combined for the IQ estimate and performed the Berlin Emotion Recognition Test (BERT)[8].

Audio and video streams were time-locked by aligning them based on the sound and visual of a movie clap denoting the start of the task. Videos and audio tracks were cut to 10min length using DaVinci Resolve.

S3 Measures

We collected the following self-report questionnaires:

- Autism-like traits: Autism Quotient, AQ[9]
- Empathy: Saarbrücker Persönlichkeitsfragebogen, SPF[10], which is the German version of the interpersonal reactivity index
- Alexithymia: Toronto Alexithymia Scale, TAS20[11]
- Depressiveness: Beck Depression Inventory, BDI[12]
- Self-monitoring: Self-monitoring Scale, SMS[13]
- Movement difficulties: Adult Dyspraxia Checklist, ADC[14]

For all participants recruited for this project, we also collected the short version of the Borderline Symptom List, BSL-23[15].

For 37 of the dyads, we used a plexiglass screen placed between the interaction partners on the table to decrease the chances of spreading an undetected infection. We applied a transparent anti-reflection foil to reduce any mirroring effects. During the conversation, participants took off their face masks. We asked these participants how much the plexiglass influenced them during the interactions (plexi; scale from 0 to 3). All participants were asked how much the cameras influenced their behaviour (video; scale from 0 to 3). Rapport is a sum of the following ratings, all on scales from 0 to 3, thus ranging from 0 to 15:

- How likeable was your interaction partner?
- How friendly was your interaction partner?
- How comfortable did you feel during the conversation?
- How smooth was the communication between you and your conversation partner?
- How well did your conversation partner respond to you?

Table S1: Group comparisons

measurement	ASD	BPD	COMP	BPD vs. ASD	COMP vs. ASD	COMP vs. BPD
ADC_total	49.06 (±17.17), n = 17	76.95 (±14.98), n = 21	34.99 (±23.86), n = 82	0.006*	0.243	0.000*
AQ_total	34.29 (±6.18), n = 17	23.67 (±4.72), n = 21	14.57 (±5.27), n = 82	0.000*	0.000*	0.000*
BDI_total	15.94 (±11.57), n = 17	24.81 (±10.48), n = 21	3.99 (±3.79), n = 82	0.715	0.000*	0.000*
BERT.acc	0.80 (±0.07), n = 17	0.82 (±0.09), n = 21	0.83 (±0.07), n = 82	1.000	0.919	1.000
BERT.rt	5.79 (±2.78), n = 17	3.84 (±1.24), n = 21	3.33 (±1.33), n = 82	0.453	0.000*	0.637
BSL_total	NaN (±NA), n = 0	46.71 (±21.44), n = 21	7.19 (±6.84), n = 37	NA	NA	0.000*
IQ.estimate	116.47 (±13.90), n = 17	109.60 (±9.54), n = 20	112.77 (±13.26), n = 82	1.000	1.000	1.000
SMS_total	5.47 (±2.81), n = 17	11.67 (±2.11), n = 21	9.80 (±2.62), n = 82	0.000*	0.000*	0.027*
SPF_total	37.00 (±7.20), n = 17	40.95 (±8.36), n = 21	44.73 (±5.82), n = 82	1.000	0.000*	0.328
TAS_total	61.76 (±11.03), n = 17	55.33 (±11.60), n = 21	37.55 (±8.53), n = 82	1.000	0.000*	0.000*
age	37.59 (±13.19), n = 17	28.38 (±10.02), n = 21	28.89 (±10.18), n = 82	0.506	0.243	1.000
plexi	0.82 (±0.53), n = 17	0.85 (±0.90), n = 13	0.86 (±0.60), n = 43	1.000	1.000	1.000
rapport	12.24 (±2.19), n = 17	11.95 (±2.64), n = 21	12.33 (±2.39), n = 81	1.000	1.000	1.000
video	0.71 (±0.69), n = 17	0.81 (±0.75), n = 21	0.63 (±0.68), n = 82	1.000	1.000	1.000

S3.1 Medication and clinical presentation

Of our clinical participants, 69.2% were taking psychotropic medication, most commonly antidepressants (ASD: 41.2%; BPD: 72.7%), antipsychotics (ASD: 11.8%; BPD: 40.9%), anticonvulsants (ASD: 5.9%; BPD: 22.7%) or a combination thereof. Thus, our sample seems clinically realistic in its heterogeneity of medication intake[16–18]. We used the Autism Quotient (AQ)[9] and the Borderline Symptom List 23 (BSL-23)[15] to further characterise our clinical samples. In our ASD sample, the AQ values were on average 34.3 ± 6.2 points, with 64.7% autistic adults scoring above the threshold of 32 points[9]. In our BPD sample, 9.5% showed extremely high, 9.5% very high, 38.1% high, 23.8% moderate and 19% mild symptoms according to the BSL-23[19].

S3.2 Group comparisons based on diagnostic status

Participants did not differ in age (ASD: $\bar{x}$ = 37.59 ± 13.19; BPD: $\bar{x}$ = 28.38 ± 10.02; COMP: $\bar{x}$ = 28.89 ± 10.18; BPD vs. ASD: pFDR = 0.506; ASD vs. COMP: pFDR = 0.243; BPD vs. COMP: pFDR = 1), IQ estimate (ASD: $\bar{x}$ = 116.47 ± 13.90; BPD: $\bar{x}$ = 109.60 ± 9.54; COMP: $\bar{x}$ = 112.77 ± 13.26; each pFDR = 1), but there was a trend for a difference in gender distribution (ASD: 6 female, 11 male; BPD: 14 female, 7 male; COMP: 52 female, 30 male; p = 0.078). However, our exploration of the influence of gender on decision scores in our support vector machines revealed no difference. Testing duration was two to three and a half hours.

```
##       ilabel
## gender ASD BPD COMP
##    fem   6  14   52
##    mal  11   7   30
```

```
##
##  Pearson's Chi-squared test
##
## data:  tb.gen
## X-squared = 5.1108, df = 2, p-value = 0.07766
```

S4 Features

We extracted head position (each three translational and three rotational parameters) and facial expressions as the strength of the activation of action units using OpenFace 2.0[20]. We used Motion Energy Analysis (MEA) to capture motion quantity for head and body regions from the scene camera[7]. Audio data was analysed using a custom processing pipeline in praat[21] which incorporated the uhm-o-meter by De Jong and colleagues[3,4]. This pipeline extracted individual phonetic and turn-taking features. Furthermore, we computed synchrony measures using windowed cross-lagged correlations as implemented in the rMEA package[22].

S4.1 Facial expressions extracted from OpenFace

We only included data of participants with a mean confidence of tracked frames greater than 75% and more than 90% successfully tracked frames. Facial expressions were captured as action units. We did not extract emotional expressions from these facial expressions as coherence between facial expressions and emotions is not a given and might be even less so for autistic people[23].

For the calculation of synchronisation, we included rotational parameters (yaw, roll, pitch) as well as the same action units as in our previous study[1]:

- Mealplanning: 1, 2, 6, 7, 9, 14, 15, 17, 20, 25, 26 and 45
- Hobbies: 1, 2, 6, 7, 9, 15, 17, 20, 23, 25, 26 and 45

We also extracted total facial expressiveness as mean intensity of all action units for each interaction partner to be included in the *MovEx* and the *CROSSturn* models. For the *CROSSturn* model, we also included other action units, as listed below.

These correspond to the following movements:

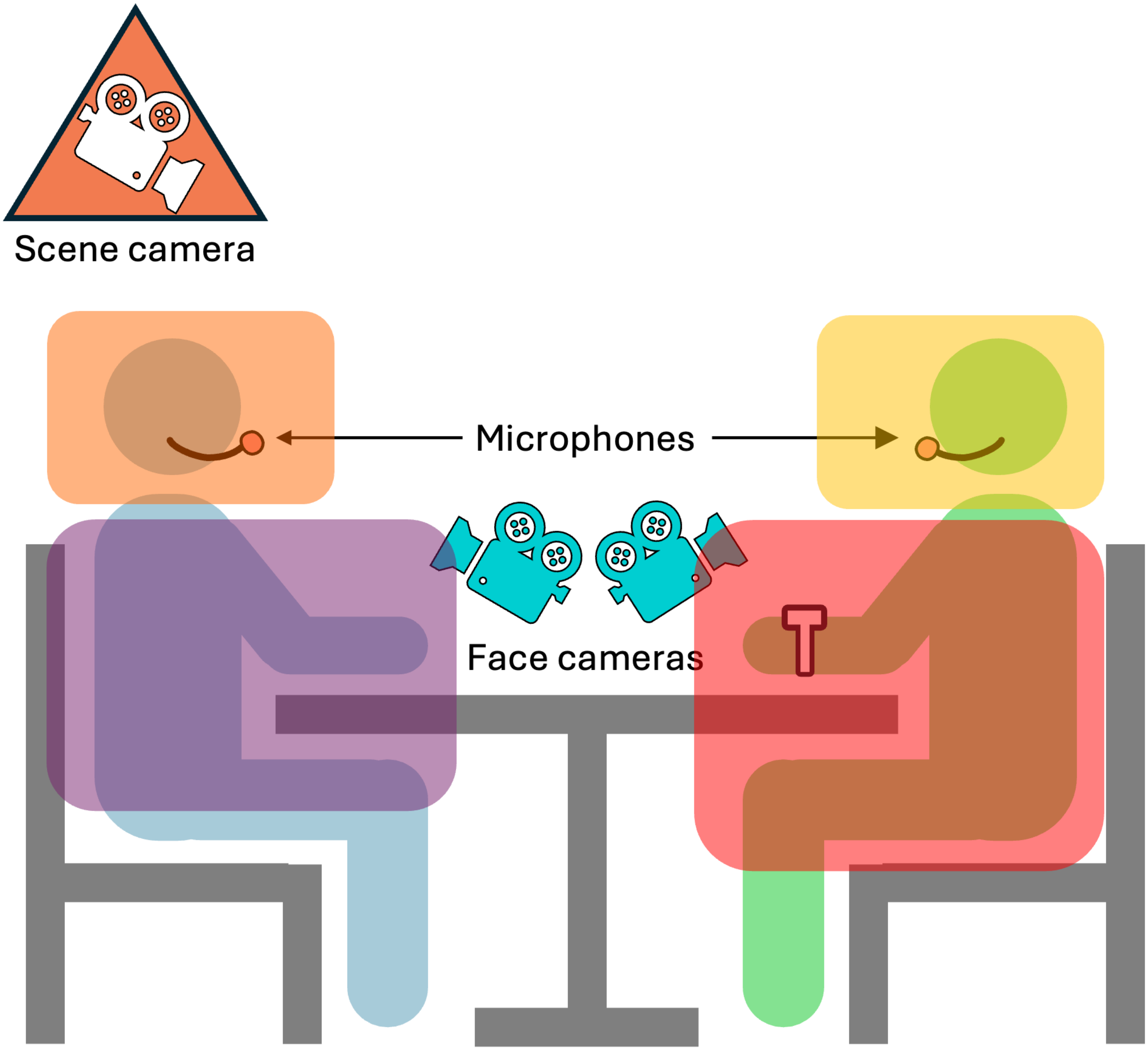
- AU1: inner brow raiser
- AU2: outer brow raiser
- AU4: brow lowerer (only *CROSSturn*)
- AU5: upper lid raiser (only *CROSSturn*)
- AU6: cheek raiser
- AU7: lid tightener
- AU9: nose wrinkler
- AU10: upper lip raiser (only *CROSSturn*)
- AU12: lip corner puller (only *CROSSturn*)
- AU14: dimpler
- AU15: lip corner depressor
- AU17: chin raiser
- AU20: lip stretcher
- AU23: lip tightener
- AU25: lips part
- AU26: jaw drop
- AU45: blink

Furthermore, we used translational head position parameters to infer head motion using the following formula with Δ_t referring to the respective frame-to-frame changes:

$$\text{head movement} = \sqrt{\Delta_t x^2 + \Delta_t y^2 + \Delta_t z^2}$$

S4.2 Motion quantity extracted using Motion Energy Analysis

This figure shows body (red and purple) and head (yellow and orange) regions of interests of each interaction partner separately:



There was always a space between the head and body region. Regions were chosen such that they cover the full range of motion throughout one conversation of one interaction partner. Thus, their sizes differed which is why we scaled all values.

In addition to using the motion quantity to compute synchronisation, we also extracted total movement in each region of interest for each interaction partner to be included in the *MovEx* model.

S4.3 Speech and turn-taking features

We extracted pitch using praat's autocorrelation method, a technique widely recognized for its reliability and accuracy[21]. We implemented a two-step pitch extraction method, as outlined by Hirst[24]. First, to capture a broad range of frequencies, we set a low pitch floor of 50 Hz and a high pitch ceiling of 700 Hz, with a time step of 15 ms. All other parameters were set to Praat's default values. Second, using these initial pitch values, we determined the first and third quartiles of pitch for each participant and task. We then used these quartiles to compute individual pitch floors and ceilings with the following algorithm:

$$\text{floor} = \min\left(0.75 \cdot Q_{1,hobbies}, 0.75 \cdot Q_{1,mealplanning}\right)$$

$$\text{ceiling} = \max\left(2.5 \cdot Q_{3,hobbies}, 2.5 \cdot Q_{3,mealplanning}\right)$$

We then used these individual pitch floors (range = 46 to 168Hz, mean = 109.7 ± 31.8) and ceilings (range = 250 to 839Hz, mean = 481.7 ± 147.4) to extract pitch. To ensure an equal number of frames for all participants, we maintained a consistent time step across all analyses. By default, praat calculates this time step using the following formula:

$$\text{timestep} = \frac{0.75}{\text{floor}}$$

Here, we used the same time step of 0.016 as in our previous study[2] which was determined based on the minimum individual pitch floor of that sample. Since the new sample did not include anyone with a lower pitch floor, the time step fits both samples.

Intensity was extracted by convolving the squared sound with a Gaussian analysis window. We used praat's default values of minimum pitch 100Hz and time step of 0.01s.

To estimate synchrony, we extracted continuous pitch and intensity time series for every millisecond of the recording. For pitch extraction, we used consistent parameters across all participants instead of individualized settings. This was necessary because the analysis width depends on the pitch floor. Given the heterogeneity of our sample, we opted for a wide range of considered frequencies, setting the pitch floor at 50Hz and the pitch ceiling at 700Hz. For intensity, we relied on Praat's default values.

In the case of turn-based synchronization, we correlated the median pitch or intensity of each turn with the median pitch or intensity of the preceding turn.

Next, we used the uhm-o-meter[3,4] to differentiate between periods of speaking and silence, identify syllables and extract several prosodic features (total number of syllables, total number of silent phases, duration of speaking as phonation time, speech rate as number of syllables per second, articulation rate as number of syllables per phonation time, average syllable duration and silence-to-turn ratio). The resulting speaking and silent instances were visually and aurally inspected to verify the accuracy of the algorithm.

S4.4 Cross-modal features

We captured two types of cross-modal features:

- Interpersonal synchronisation of one person's head with the other person's body movement and vice versa
- AU activation, body and head movement during listening and speaking

S4.5 Synchrony computations

We used the following settings for our windowed lagged cross-correlation (WLCC): For lag, step and window size, we stayed in line with previous literature[28]. We applied Fisher's Z-transformation and converted all values to their absolute[29]. Furthermore, we computed turn-based adaptation of pitch and intensity.

Table S2: WLCC settings in seconds

measure	window	step	lag
Facial action units synchronisation	7	4	2
Body MEA synchronisation	30	15	5
Head MEA synchronisation	30	15	5
Intrapersonal synchrony	30	15	5
Pitch synchrony	16	8	2
Intensity synchrony	16	8	2

For each window, the maximum correlation value was chosen out of all relevant lags (peack-picking). We cross-correlated head movements (from OpenFace) with body motion energy time series (from MEA) to estimate intrapersonal synchrony.

S4.6 Feature lists

This list shows all features without the information of conversation, i.e., each of these features was added twice to the model, once from the mealplanning and once from the hobbies conversation. The number of features per model is displayed as well. For many of the extracted features, we calculated summary scores some which are indicated by abbreviations (mean, md = median, sd = standard deviation, min = minimum, max = maximum, ske = skewness and kurtosis)

Table S3: List of all mealplanning features of the base models

model	features
BODYsync (14 features)	min_M_bodysync, max_M_bodysync, sd_M_bodysync, mean_M_bodysync, md_M_bodysync, skew_M_bodysync, kurtosis_M_bodysync
CROSSsync (28 features)	min_M_LOF, min_M_ROF, max_M_LOF, max_M_ROF, md_M_LOF, md_M_ROF, mean_M_LOF, mean_M_ROF, sd_M_LOF, sd_M_ROF, kurtosis_M_LOF, kurtosis_M_ROF, skew_M_LOF, skew_M_ROF

CROSSturn (532 features)

self_min_M_AU01_r, self_min_M_AU02_r, self_min_M_AU04_r, self_min_M_AU05_r,
self_min_M_AU06_r, self_min_M_AU07_r, self_min_M_AU09_r, self_min_M_AU10_r,
self_min_M_AU12_r, self_min_M_AU14_r, self_min_M_AU15_r, self_min_M_AU17_r,
self_min_M_AU20_r, self_min_M_AU23_r, self_min_M_AU25_r, self_min_M_AU26_r,
self_min_M_AU45_r, self_min_M_MEA_body, self_min_M_MEA_head, self_max_M_AU01_r,
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other_skew_M_AU20_r, other_skew_M_AU23_r, other_skew_M_AU25_r, other_skew_M_AU26_r,
other_skew_M_AU45_r, other_skew_M_MEA_body, other_skew_M_MEA_head

FACEsync (168 features)	min_M_AU01_r, max_M_AU01_r, sd_M_AU01_r, mean_M_AU01_r, md_M_AU01_r, skew_M_AU01_r, kurtosis_M_AU01_r, min_M_AU02_r, max_M_AU02_r, sd_M_AU02_r, mean_M_AU02_r, md_M_AU02_r, skew_M_AU02_r, kurtosis_M_AU02_r, min_M_AU06_r, max_M_AU06_r, sd_M_AU06_r, mean_M_AU06_r, md_M_AU06_r, skew_M_AU06_r, kurtosis_M_AU06_r, min_M_AU07_r, max_M_AU07_r, sd_M_AU07_r, mean_M_AU07_r, md_M_AU07_r, skew_M_AU07_r, kurtosis_M_AU07_r, min_M_AU09_r, max_M_AU09_r, sd_M_AU09_r, mean_M_AU09_r, md_M_AU09_r, skew_M_AU09_r, kurtosis_M_AU09_r, min_M_AU14_r, max_M_AU14_r, sd_M_AU14_r, mean_M_AU14_r, md_M_AU14_r, skew_M_AU14_r, kurtosis_M_AU14_r, min_M_AU15_r, max_M_AU15_r, sd_M_AU15_r, mean_M_AU15_r, md_M_AU15_r, skew_M_AU15_r, kurtosis_M_AU15_r, min_M_AU17_r, max_M_AU17_r, sd_M_AU17_r, mean_M_AU17_r, md_M_AU17_r, skew_M_AU17_r, kurtosis_M_AU17_r, min_M_AU20_r, max_M_AU20_r, sd_M_AU20_r, mean_M_AU20_r, md_M_AU20_r, skew_M_AU20_r, kurtosis_M_AU20_r, min_M_AU25_r, max_M_AU25_r, sd_M_AU25_r, mean_M_AU25_r, md_M_AU25_r, skew_M_AU25_r, kurtosis_M_AU25_r, min_M_AU26_r, max_M_AU26_r, sd_M_AU26_r, mean_M_AU26_r, md_M_AU26_r, skew_M_AU26_r, kurtosis_M_AU26_r, min_M_AU45_r, max_M_AU45_r, sd_M_AU45_r, mean_M_AU45_r, md_M_AU45_r, skew_M_AU45_r, kurtosis_M_AU45_r
HEADsync (56 features)	min_M_headsync, max_M_headsync, sd_M_headsync, mean_M_headsync, md_M_headsync, skew_M_headsync, kurtosis_M_headsync, min_M_pose_Rxsync, max_M_pose_Rxsync, sd_M_pose_Rxsync, mean_M_pose_Rxsync, md_M_pose_Rxsync, skew_M_pose_Rxsync, kurtosis_M_pose_Rxsync, min_M_pose_Rysync, max_M_pose_Rysync, sd_M_pose_Rysync, mean_M_pose_Rysync, md_M_pose_Rysync, skew_M_pose_Rysync, kurtosis_M_pose_Rysync, min_M_pose_Rzsync, max_M_pose_Rzsync, sd_M_pose_Rzsync, mean_M_pose_Rzsync, md_M_pose_Rzsync, skew_M_pose_Rzsync, kurtosis_M_pose_Rzsync
INTRAsync (14 features)	min_M_intra, max_M_intra, sd_M_intra, mean_M_intra, md_M_intra, skew_M_intra, kurtosis_M_intra
MovEx (6 features)	M_body_total_movement, M_head_total_movement, mean_intensity_M
Speech (30 features)	dyad_pit_sync_MEA_M_speech, dyad_int_sync_MEA_M_speech, dyad_spr_M_speech, dyad_str_M_speech, dyad_ttg_M_speech, dyad_no_turns_M_speech, nsyll_M_speech, npause_M_speech, pho_M_speech, art_M_speech, pit_sync_M_speech, int_sync_M_speech, art_sync_M_speech, pit_var_M_speech, int_var_M_speech

S5 Model performance

S5.1 Distinguishing ASD-involved from comparison interactions

All base models except the *HEADsync* and the *INTRAsync* models performed significantly better than chance according to permutation tests (*HEADsync*: $p_{FDR} = 1$; *INTRAsync*: $p_{FDR} = 1$). For this comparison, the *FACEsync* model achieved the highest balanced accuracy, specifically 74.9% (79.4% sensitivity; 70.5% specificity), closely followed by the *CROSSsync* model with 71.3% balanced accuracy (67.6% sensitivity; 75% specificity). The stacking model had a significantly higher balanced accuracy of 80.6% (79.4% sensitivity; 81.8% specificity). Here, seven ASD-involved interactions were misclassified as non-clinical, while zero non-clinical interactions were mistaken as ASD-involved interactions. The rest of the interaction partners, 63 in total, were classified accurately. The stacking did not perform significantly better than the best base model (*FACEsync*: $p_{FDR} = 0.112$) but outperformed all other base models.

S5.2 Distinguishing BPD-involved from COMP interactions

While developing an algorithm for technology-assisted diagnostics of BPD was not the explicit goal of this research project, we explored the application of our features to the classification between BPD-involved and COMP interactions. Despite the features being chosen with symptoms and characteristics of ASD in mind, the *CROSSturn*, *FACEsync*, *HEADsync* and *Speech* models performed above chance in this comparison (*BODYsync*: $p_{FDR} = 1$; *CROSSsync*: $p_{FDR} = 0.282$; *INTRAsync*: $p_{FDR} = 1$; *MovEx*: $p_{FDR} = 0.242$). Specifically, the *HEADsync* model achieved 68.7% balanced accuracy (71.4; 65.9% specificity), the *FACEsync* 64% (64.3; 65.9% specificity), the *Speech* 61.6% (59.5; 63.6% specificity) and the *CROSSturn* 55.7% (52.4; 59.1% specificity). The stacking model performed comparable to the *HEADsync* and the *MovEx* model but outperformed the other base models, reaching 65.3% balanced accuracy (73.8; 56.8% specificity). Thus, the stacking model only misclassified eleven BPD-involved interactions as COMP, but 19 COMP interactions were labelled as BPD-involved.

S5.3 One-verus-One comparisons

Table S4: Performance of all the models in One-versus-One comparisons

comparison	model	sens	spec	BAC	AUC	p.fdr	sig

ASD-COMP vs BPD-COMP	BODYsync	32.353	47.619	39.986	0.361	1.000	
ASD-COMP vs BPD-COMP	CROSSsync	52.941	69.048	60.994	0.678	0.136	
ASD-COMP vs BPD-COMP	CROSSturn	67.647	85.714	76.681	0.749	0.000	*
ASD-COMP vs BPD-COMP	FACEsync	76.471	59.524	67.997	0.719	0.000	*
ASD-COMP vs BPD-COMP	HEADsync	55.882	57.143	56.513	0.609	0.936	
ASD-COMP vs BPD-COMP	INTRAsync	38.235	54.762	46.499	0.455	1.000	
ASD-COMP vs BPD-COMP	MovEx	58.824	78.571	68.698	0.758	0.027	*
ASD-COMP vs BPD-COMP	Speech	64.706	76.190	70.448	0.771	0.000	*
ASD-COMP vs BPD-COMP	STACK	70.588	92.857	81.723	0.805	NaN	NA
ASD-COMP vs COMP-COMP	BODYsync	58.824	61.364	60.094	0.607	0.000	*
ASD-COMP vs COMP-COMP	CROSSsync	67.647	75.000	71.324	0.763	0.000	*
ASD-COMP vs COMP-COMP	CROSSturn	44.118	72.727	58.422	0.627	0.000	*
ASD-COMP vs COMP-COMP	FACEsync	79.412	70.454	74.933	0.780	0.000	*
ASD-COMP vs COMP-COMP	HEADsync	58.824	68.182	63.503	0.608	1.000	
ASD-COMP vs COMP-COMP	INTRAsync	26.471	40.909	33.690	0.297	1.000	
ASD-COMP vs COMP-COMP	MovEx	64.706	72.727	68.717	0.777	0.000	*
ASD-COMP vs COMP-COMP	Speech	64.706	72.727	68.717	0.673	0.000	*
ASD-COMP vs COMP-COMP	STACK	79.412	81.818	80.615	0.845	NaN	NA
BPD-COMP vs COMP-COMP	BODYsync	28.571	36.364	32.468	0.308	1.000	
BPD-COMP vs COMP-COMP	CROSSsync	50.000	61.364	55.682	0.594	0.282	
BPD-COMP vs COMP-COMP	CROSSturn	52.381	59.091	55.736	0.569	0.045	*
BPD-COMP vs COMP-COMP	FACEsync	64.286	63.636	63.961	0.670	0.007	*
BPD-COMP vs COMP-COMP	HEADsync	71.429	65.909	68.669	0.736	0.000	*
BPD-COMP vs COMP-COMP	INTRAsync	57.143	54.545	55.844	0.516	1.000	
BPD-COMP vs COMP-COMP	MovEx	69.048	70.454	69.751	0.686	0.242	
BPD-COMP vs COMP-COMP	Speech	59.524	63.636	61.580	0.693	0.007	*
BPD-COMP vs COMP-COMP	STACK	73.810	56.818	65.314	0.760	NaN	NA

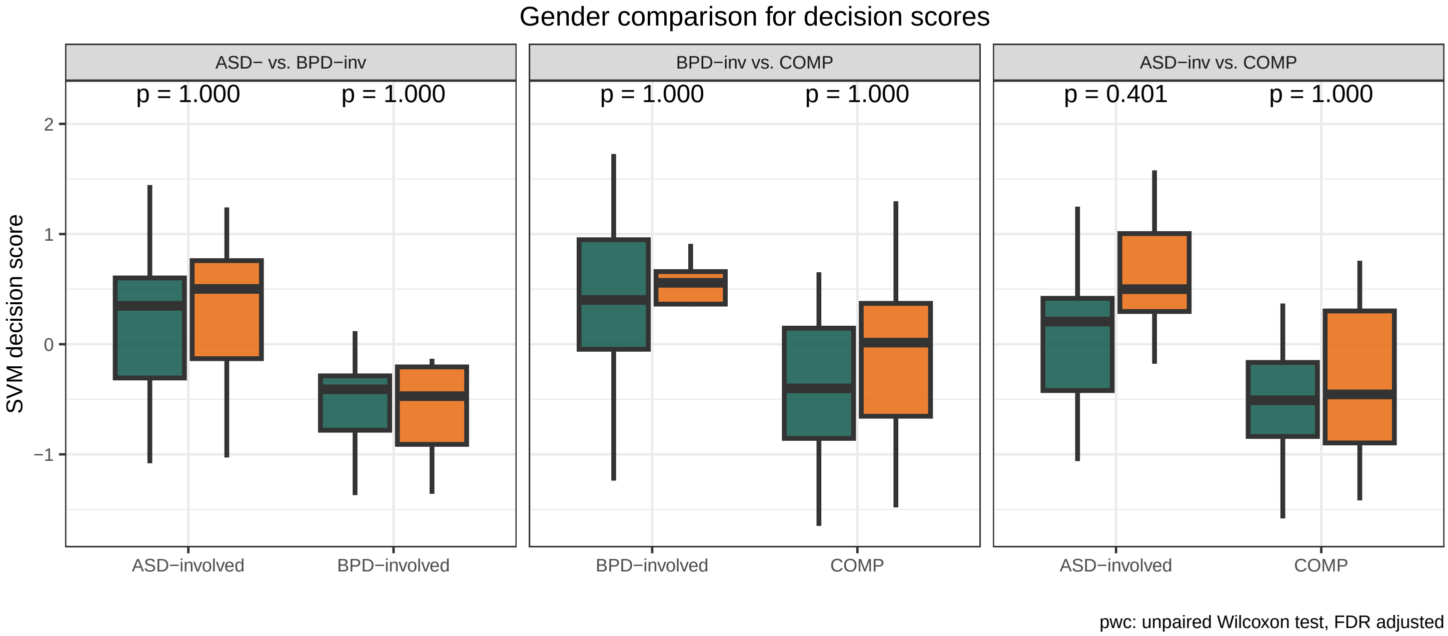
S5.4 Multi-group comparisons

Table S5: Performance of all the models in Multi-group comparisons

comparison	model	BAC	p.fdr	sig
MultiGroup	BODYsync	48.6	1.000	
MultiGroup	CROSSsync	58.1	0.000	*
MultiGroup	CROSSturn	57.3	0.000	*
MultiGroup	FACEsync	57.6	0.007	*
MultiGroup	HEADsync	54.6	0.000	*
MultiGroup	INTRAsync	49.5	1.000	
MultiGroup	MovEx	60.8	0.000	*
MultiGroup	Speech	63.9	0.000	*
MultiGroup	STACK	63.2	NaN	NA

S5.5 Gender comparisons

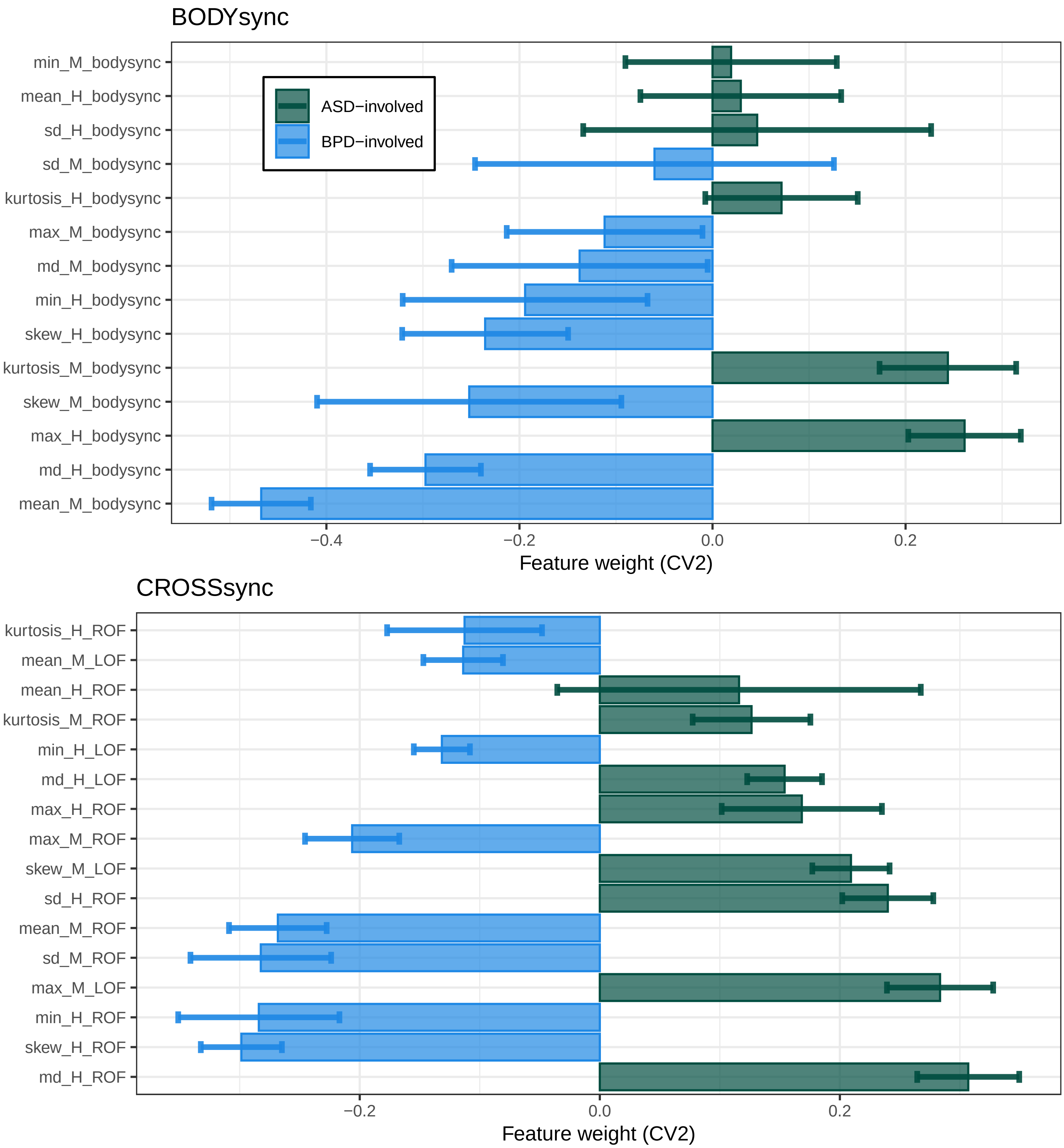
Did the models perform better for one gender than the other? We perform unpaired Wilcoxon tests for the labels and models separately.

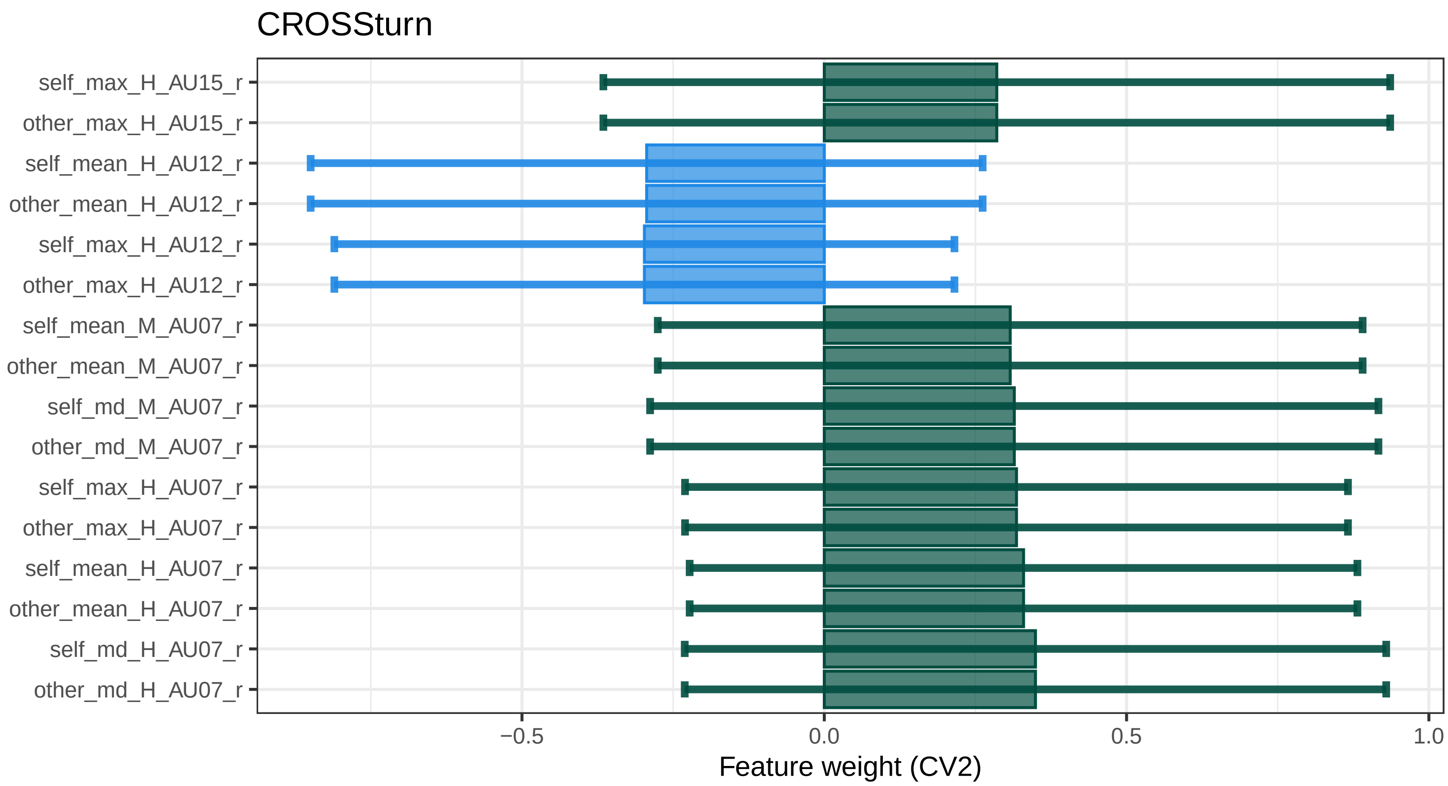


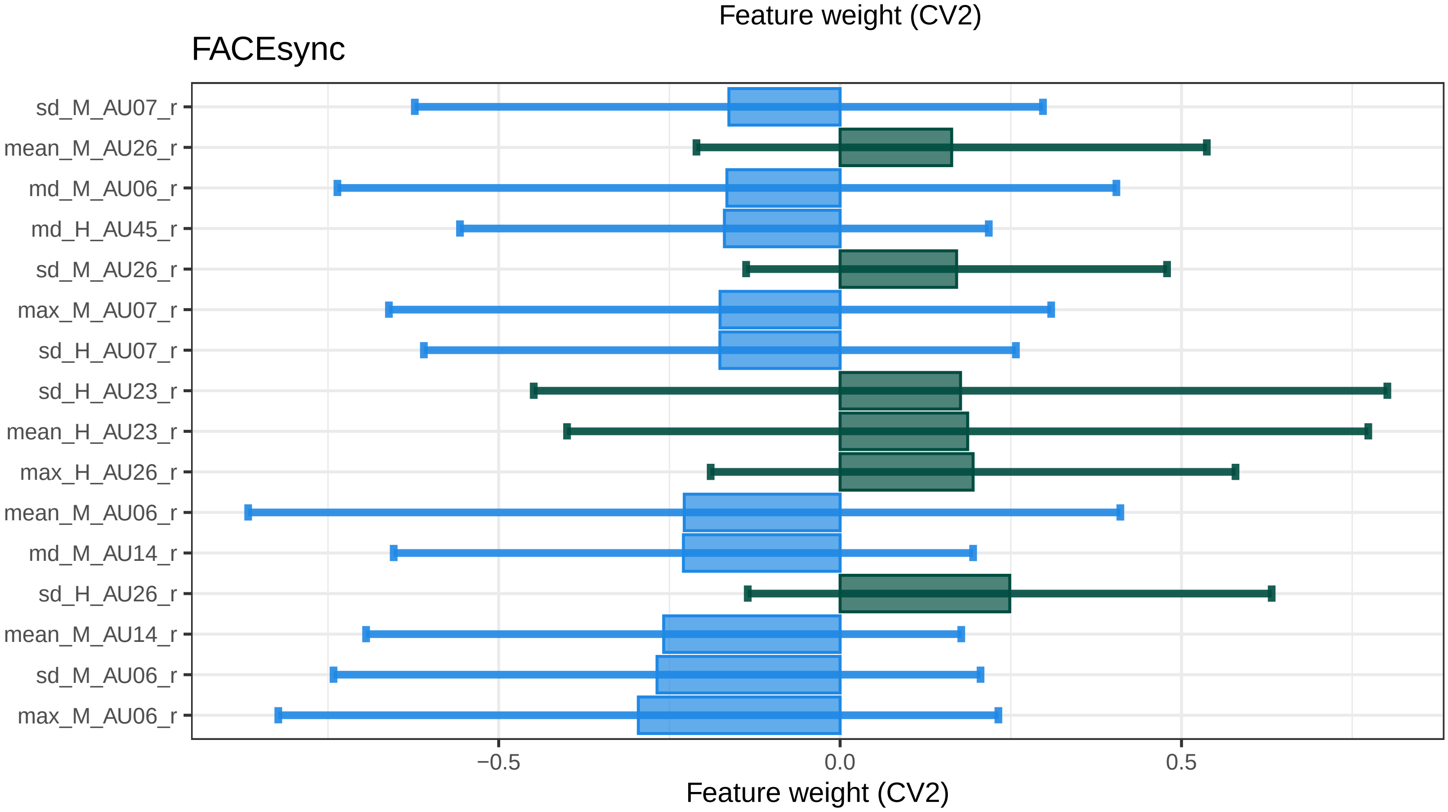
S6 Model visualisations

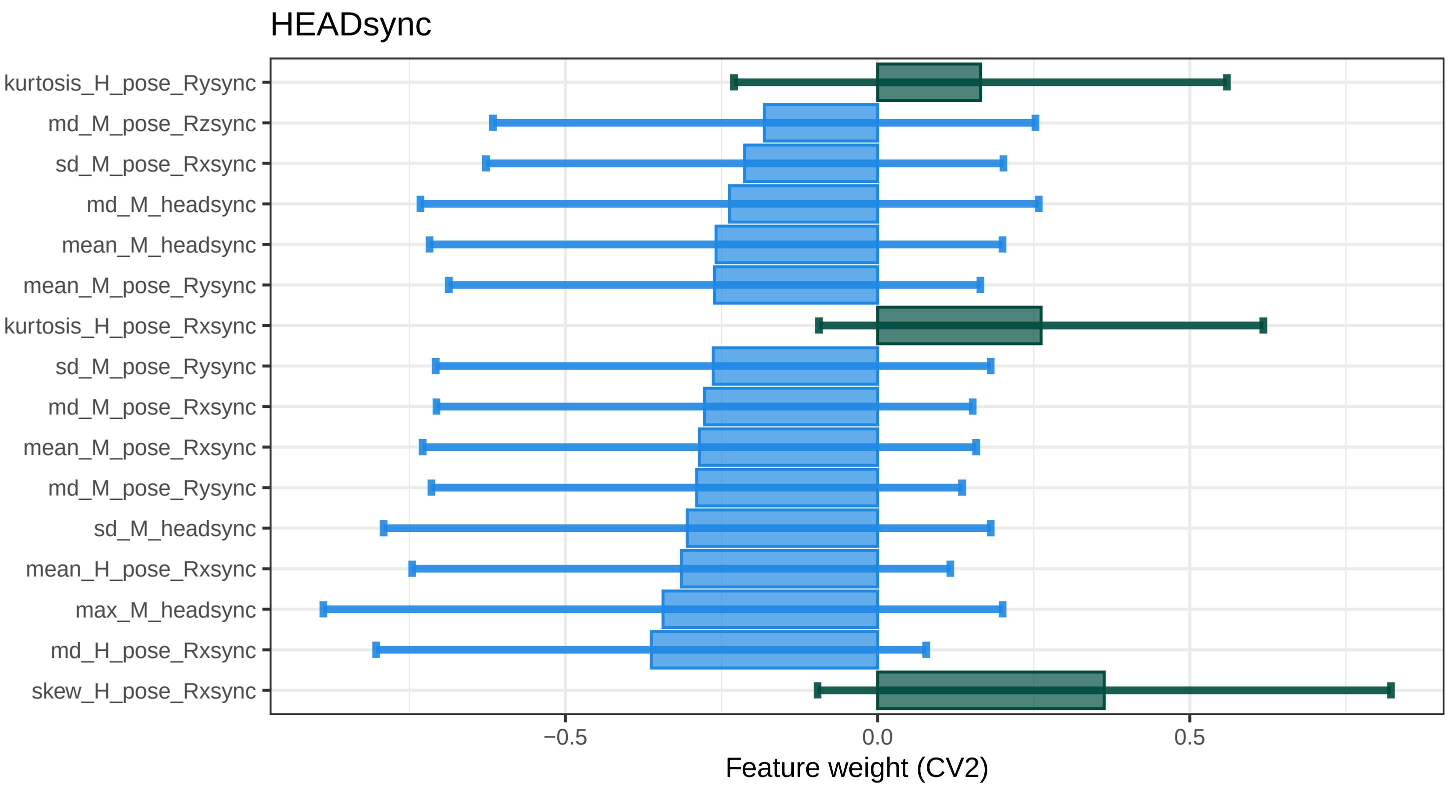
The following figures were created inspired by NeuroMiner[30] visualisations and show the sign-based consistency[31] as well as the feature weights of the models distinguishing between interaction partners from ASD- and BPD-involved interactions. For models with a large number of features, the top 16 features are plotted.

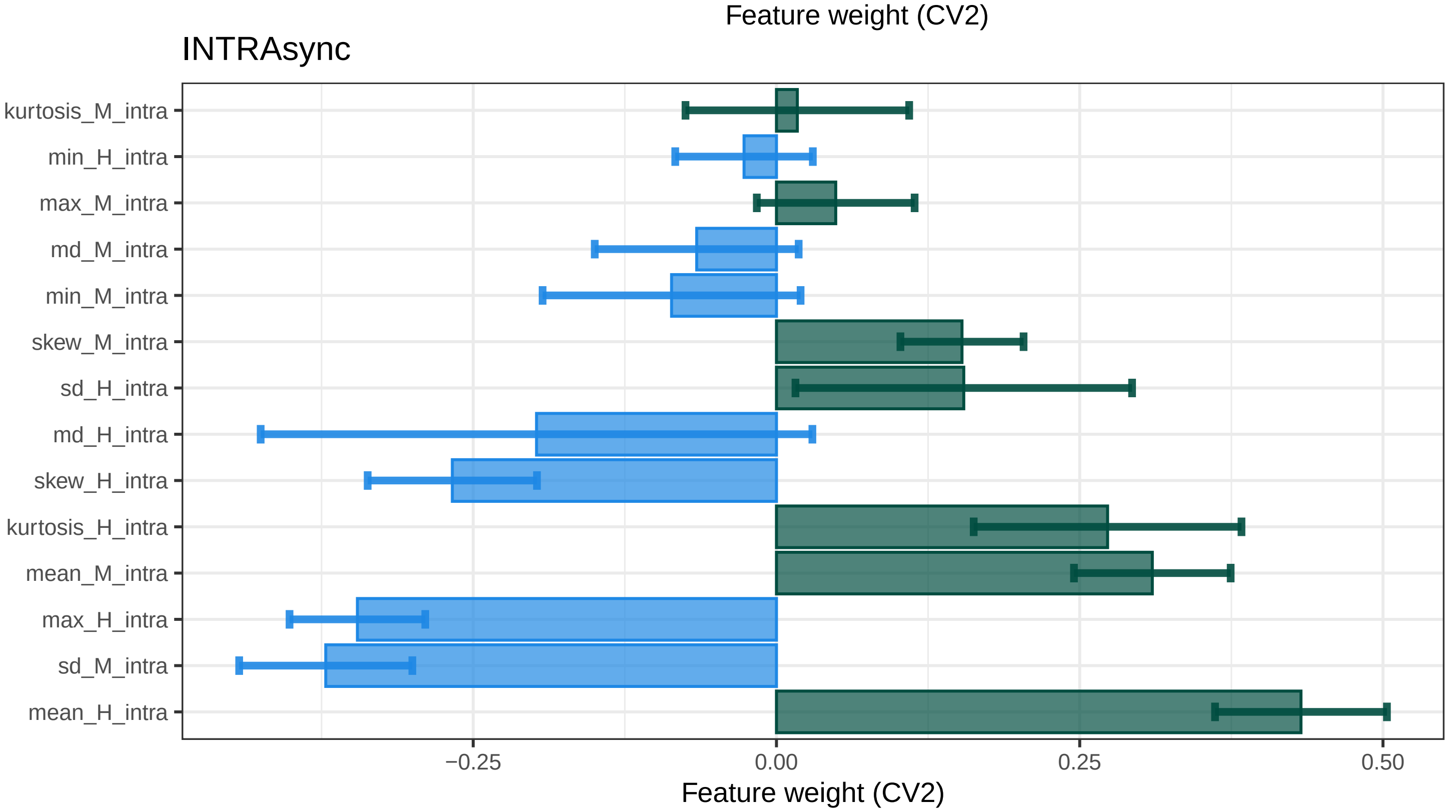
S6.1 ASD-involved versus BPD-involved classifiers

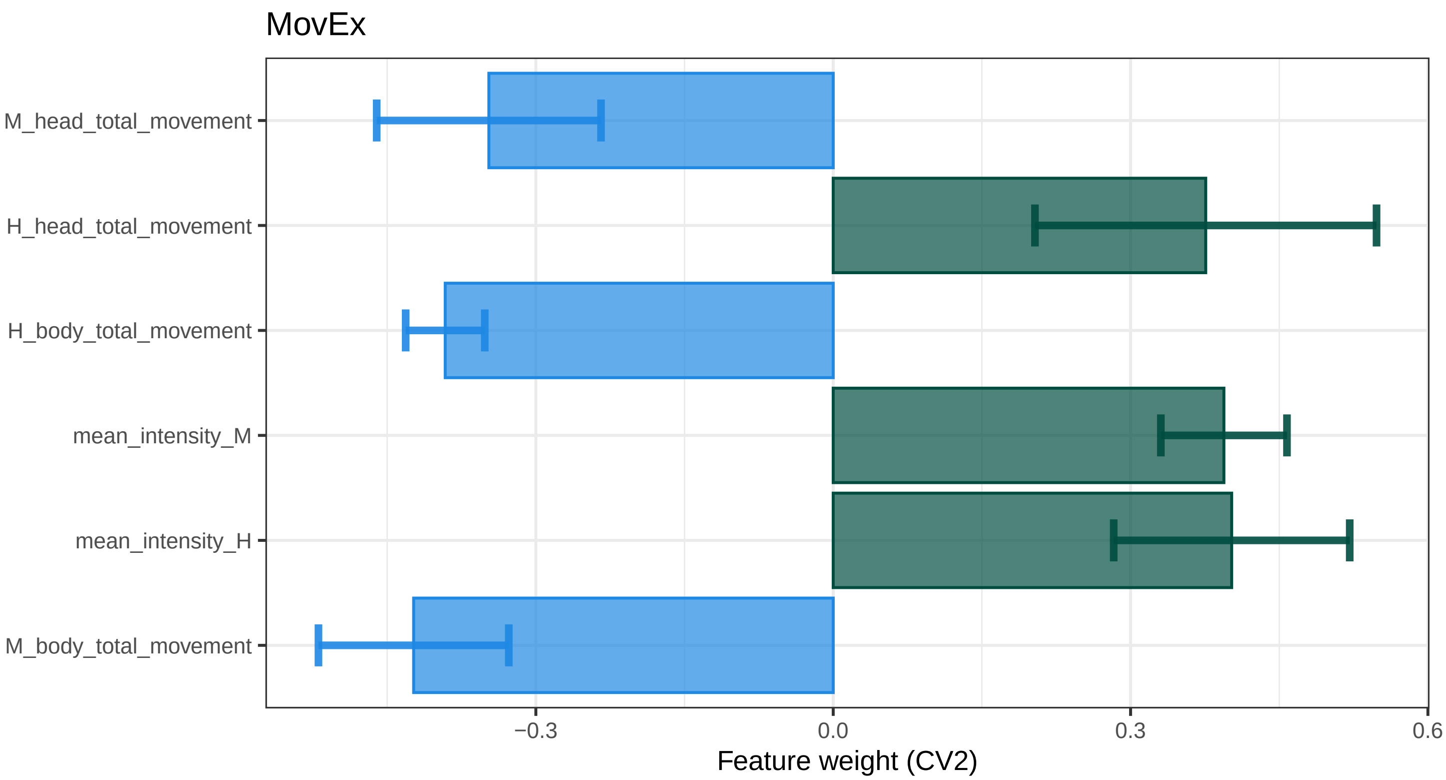


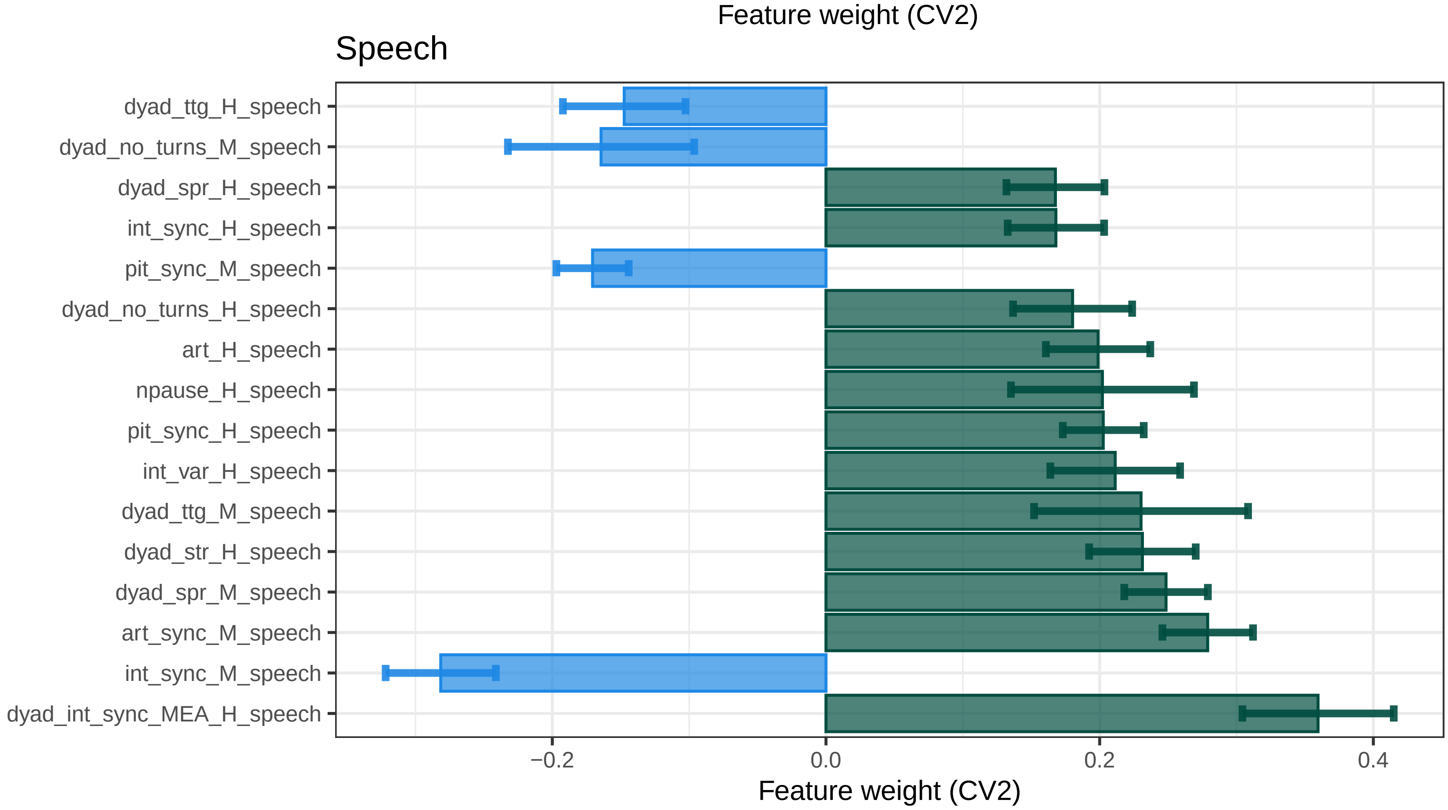

CROSSturn
self_max_H_AU15_r
other_max_H_AU15_r
self_mean_H_AU12_r
other_mean_H_AU12_r
self_max_H_AU12_r
other_max_H_AU12_r
self_mean_M_AU07_r
other_mean_M_AU07_r
self_md_M_AU07_r
other_md_M_AU07_r
self_max_H_AU07_r
other_max_H_AU07_r
self_mean_H_AU07_r
other_mean_H_AU07_r
self_md_H_AU07_r
other_md_H_AU07_r
−0.5
0.0
0.5
1.0
Feature weight (CV2)

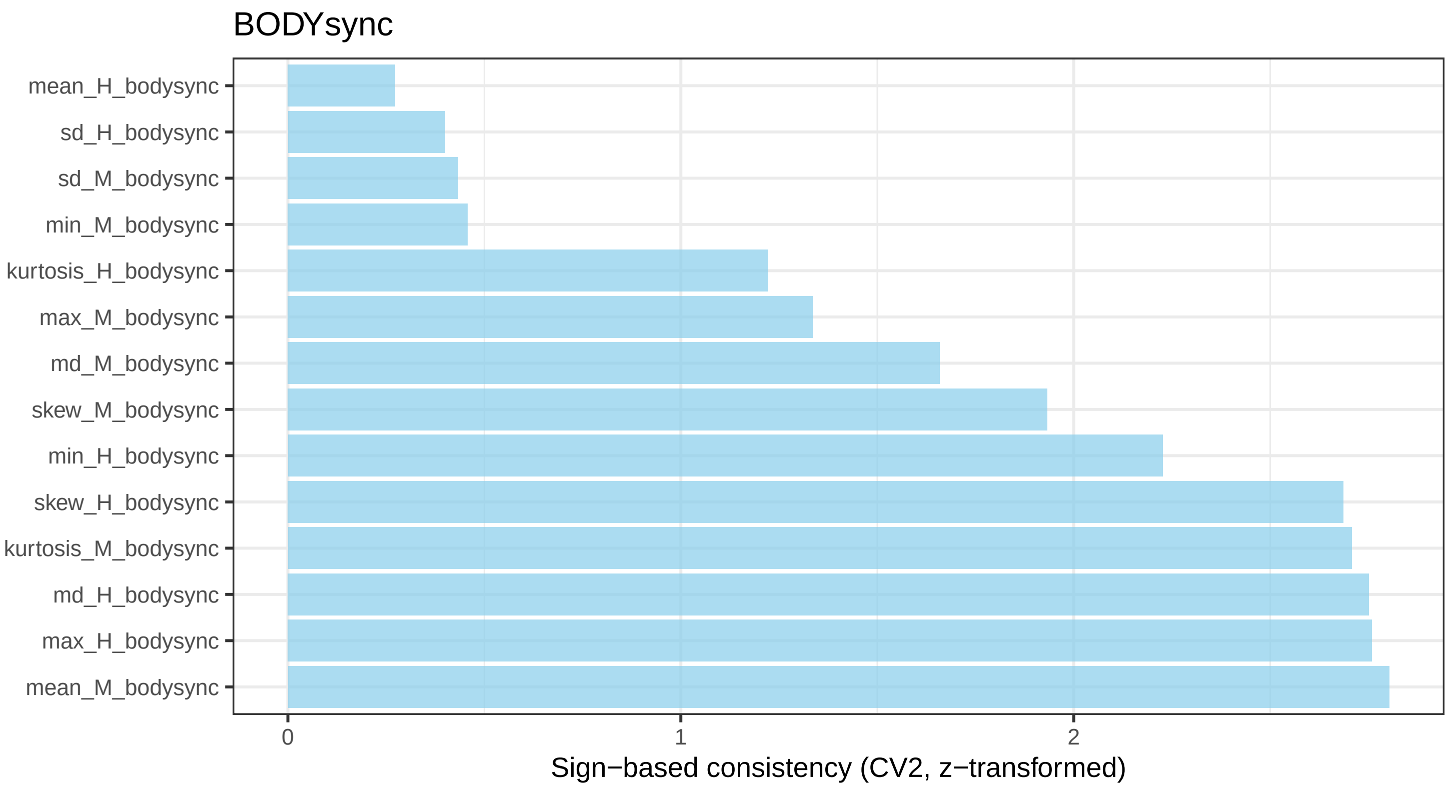

FACEsync
sd_M_AU07_r
mean_M_AU26_r
md_M_AU06_r
md_H_AU45_r
sd_M_AU26_r
max_M_AU07_r
sd_H_AU07_r
sd_H_AU23_r
mean_H_AU23_r
max_H_AU26_r
mean_M_AU06_r
md_M_AU14_r
sd_H_AU26_r
mean_M_AU14_r
sd_M_AU06_r
max_M_AU06_r
−0.5
0.0
0.5
Feature weight (CV2)

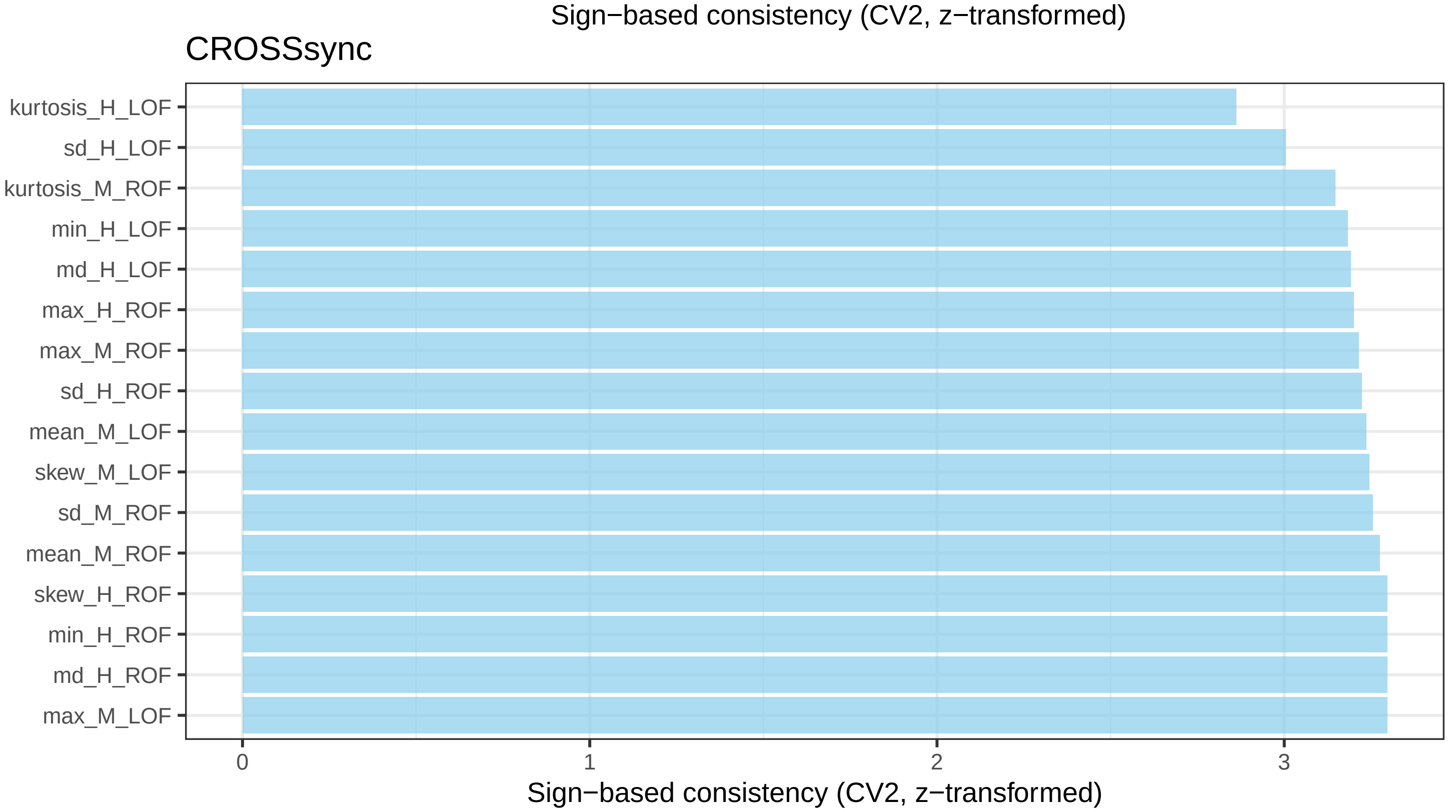

HEADsync
kurtosis_H_pose_Rysync
md_M_pose_Rzsync
sd_M_pose_Rxsync
md_M_headsync
mean_M_headsync
mean_M_pose_Rysync
kurtosis_H_pose_Rxsync
sd_M_pose_Rysync
md_M_pose_Rxsync
mean_M_pose_Rxsync
md_M_pose_Rysync
sd_M_headsync
mean_H_pose_Rxsync
max_M_headsync
md_H_pose_Rxsync
skew_H_pose_Rxsync
−0.5
0.0
0.5
Feature weight (CV2)

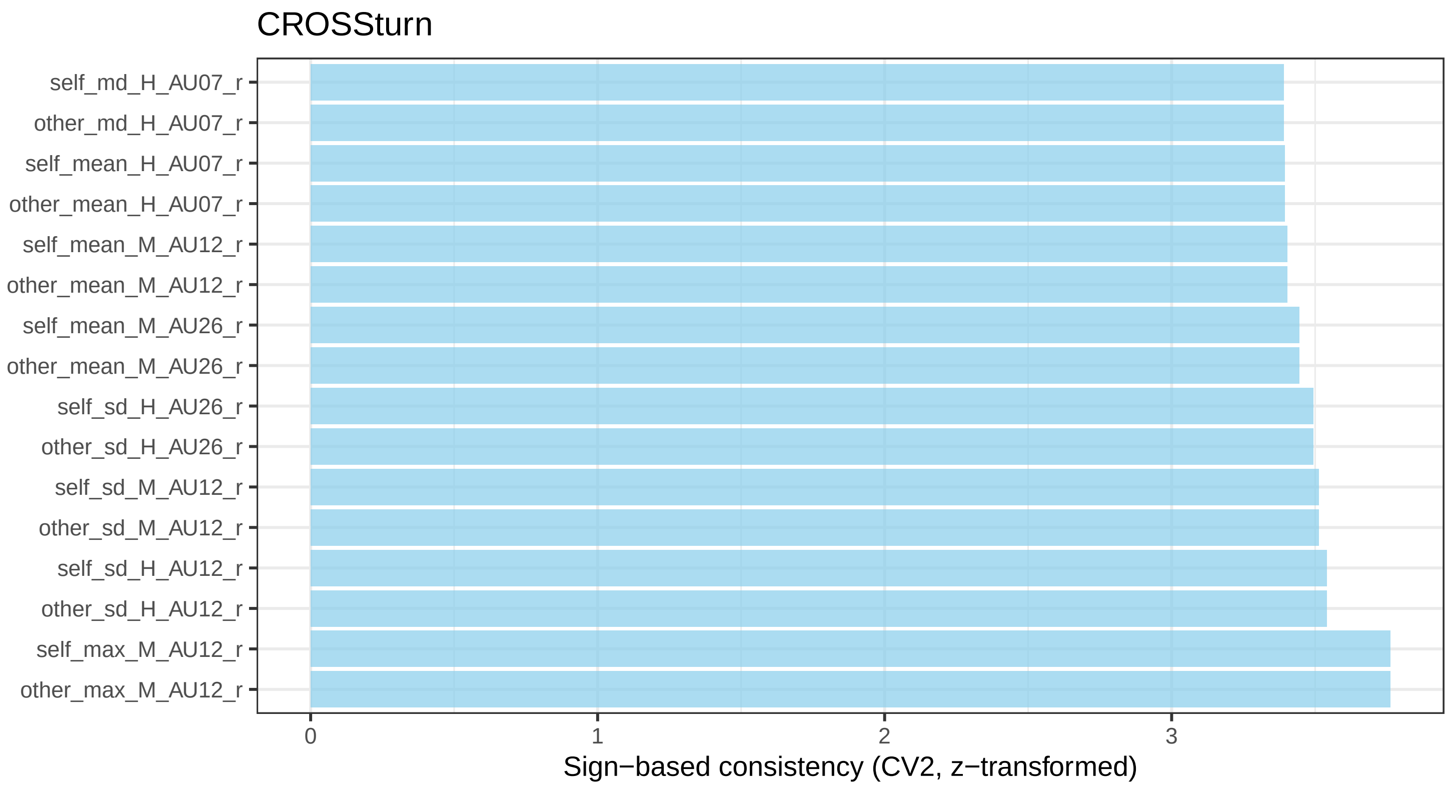

INTRAsync
kurtosis_M_intra
min_H_intra
max_M_intra
md_M_intra
min_M_intra
skew_M_intra
sd_H_intra
md_H_intra
skew_H_intra
kurtosis_H_intra
mean_M_intra
max_H_intra
sd_M_intra
mean_H_intra
−0.25
0.00
0.25
0.50
Feature weight (CV2)

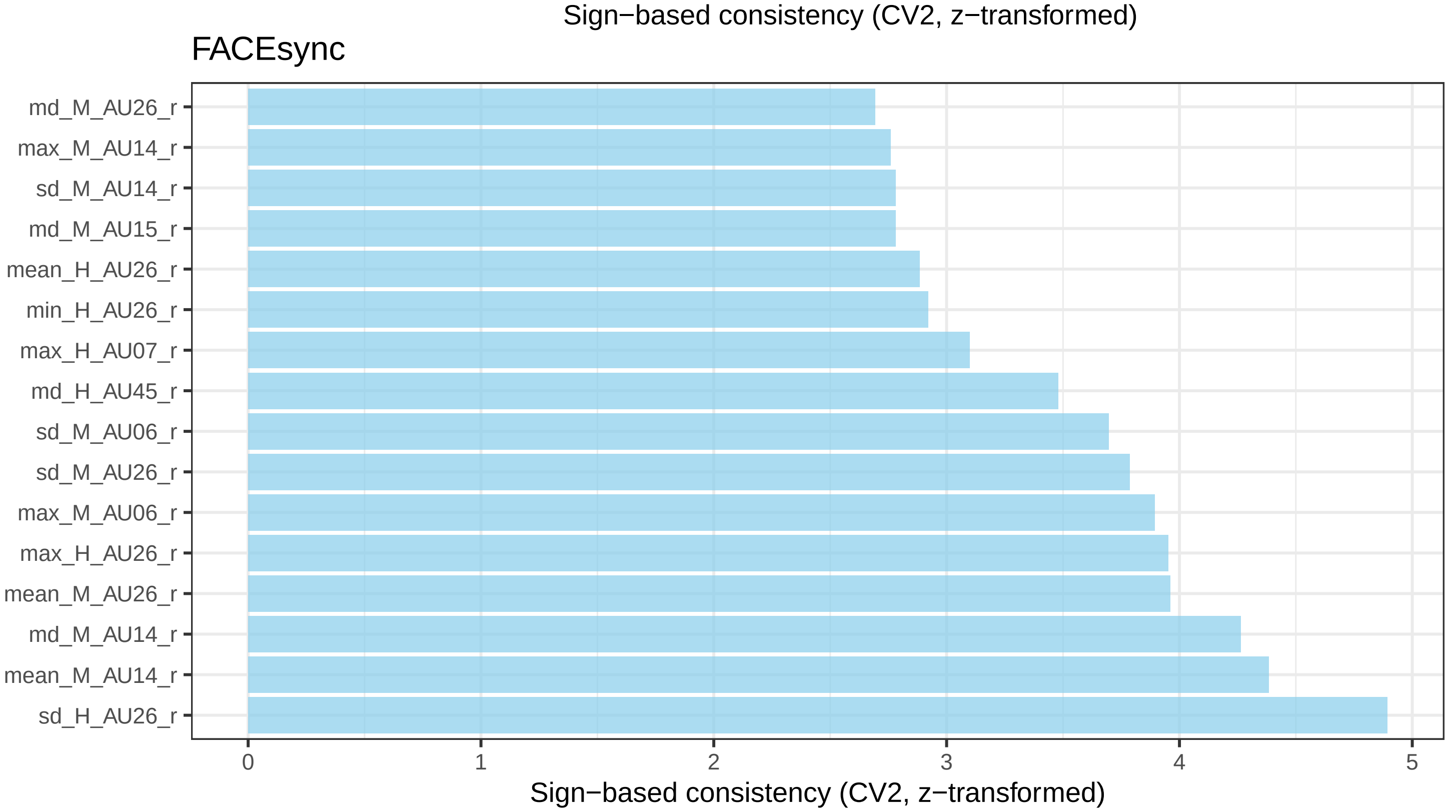

MovEx
M_head_total_movement
H_head_total_movement
H_body_total_movement
mean_intensity_M
mean_intensity_H
M_body_total_movement
−0.3
0.0
0.3
0.6
Feature weight (CV2)

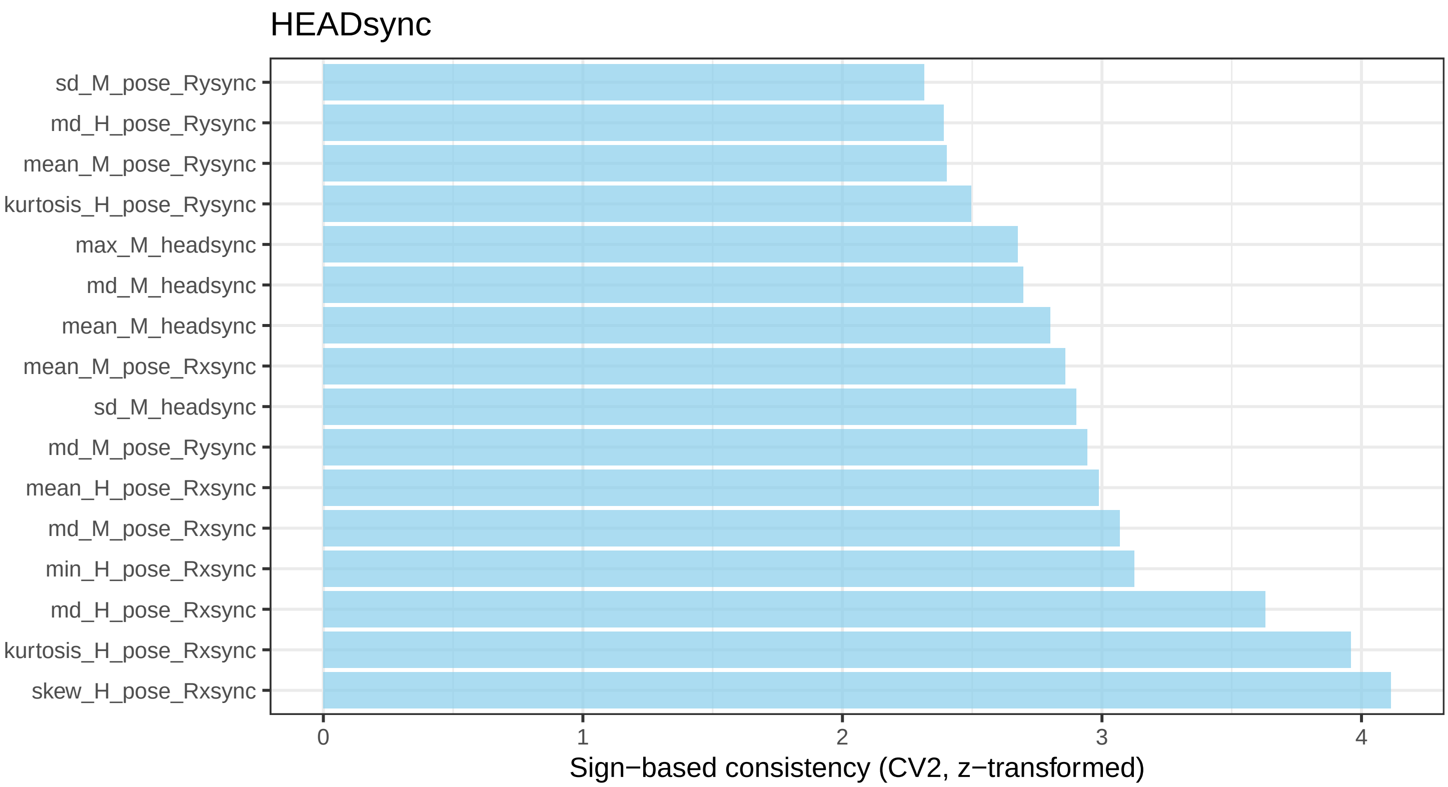

Speech
dyad_ttg_H_speech
dyad_no_turns_M_speech
dyad_spr_H_speech
int_sync_H_speech
pit_sync_M_speech
dyad_no_turns_H_speech
art_H_speech
npause_H_speech
pit_sync_H_speech
int_var_H_speech
dyad_ttg_M_speech
dyad_str_H_speech
dyad_spr_M_speech
art_sync_M_speech
int_sync_M_speech
dyad_int_sync_MEA_H_speech
−0.2
0.0
0.2
0.4
Feature weight (CV2)

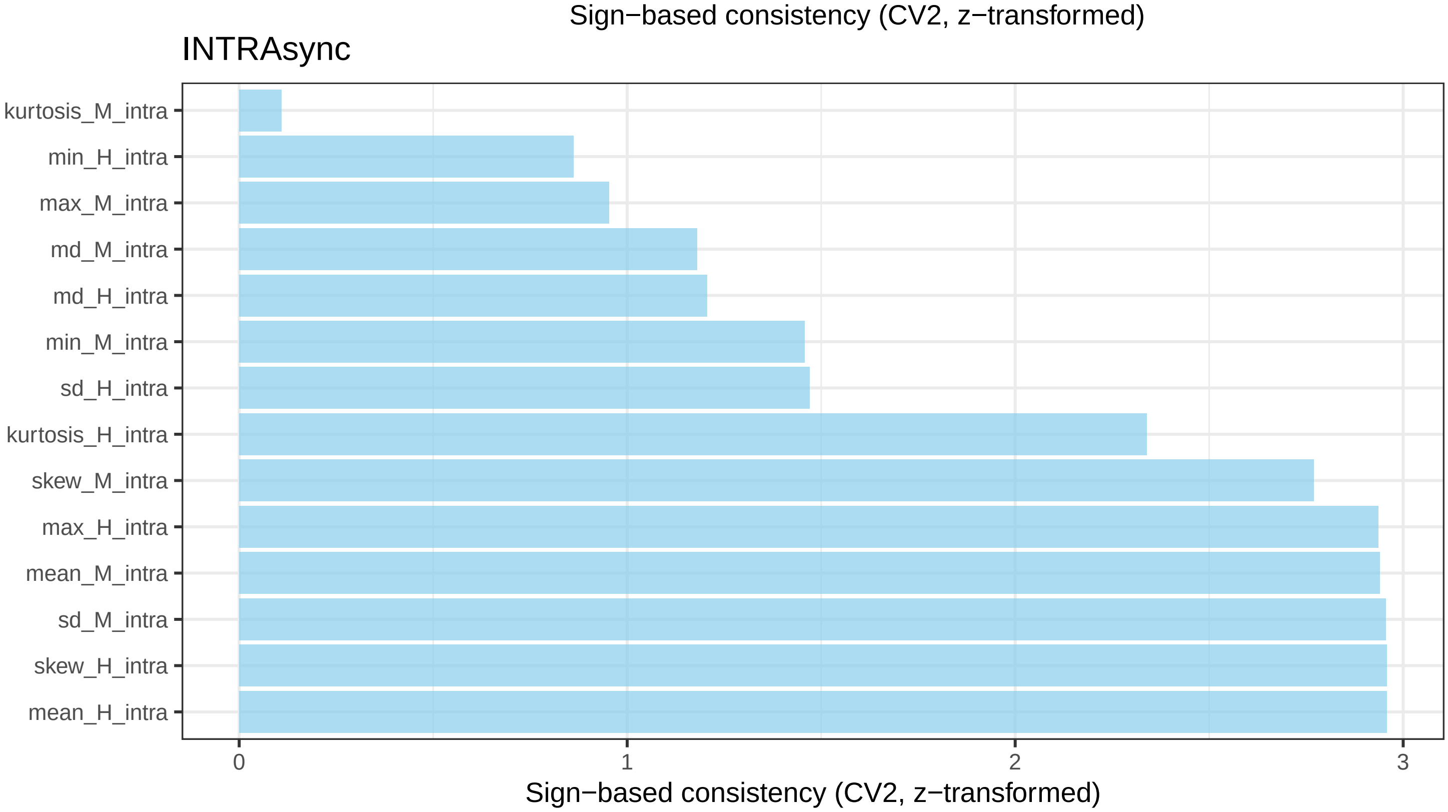

BODYsync
mean_H_bodysync
sd_H_bodysync
sd_M_bodysync
min_M_bodysync
kurtosis_H_bodysync
max_M_bodysync
md_M_bodysync
skew_M_bodysync
min_H_bodysync
skew_H_bodysync
kurtosis_M_bodysync
md_H_bodysync
max_H_bodysync
mean_M_bodysync
0
1
2
Sign−based consistency (CV2, z−transformed)

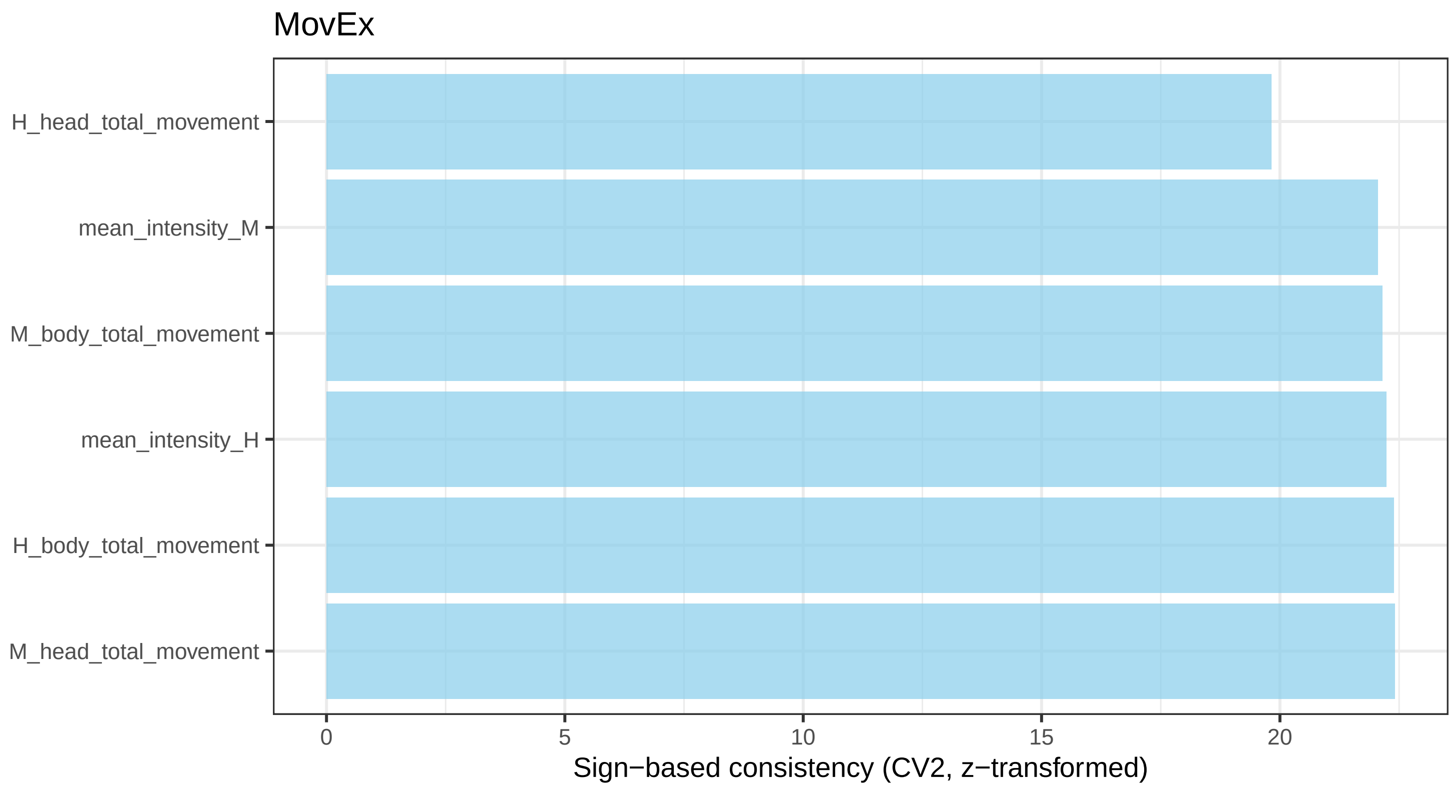

CROSSsync
kurtosis_H_LOF
sd_H_LOF
kurtosis_M_ROF
min_H_LOF
md_H_LOF
max_H_ROF
max_M_ROF
sd_H_ROF
mean_M_LOF
skew_M_LOF
sd_M_ROF
mean_M_ROF
skew_H_ROF
min_H_ROF
md_H_ROF
max_M_LOF
0
1
2
3
Sign−based consistency (CV2, z−transformed)

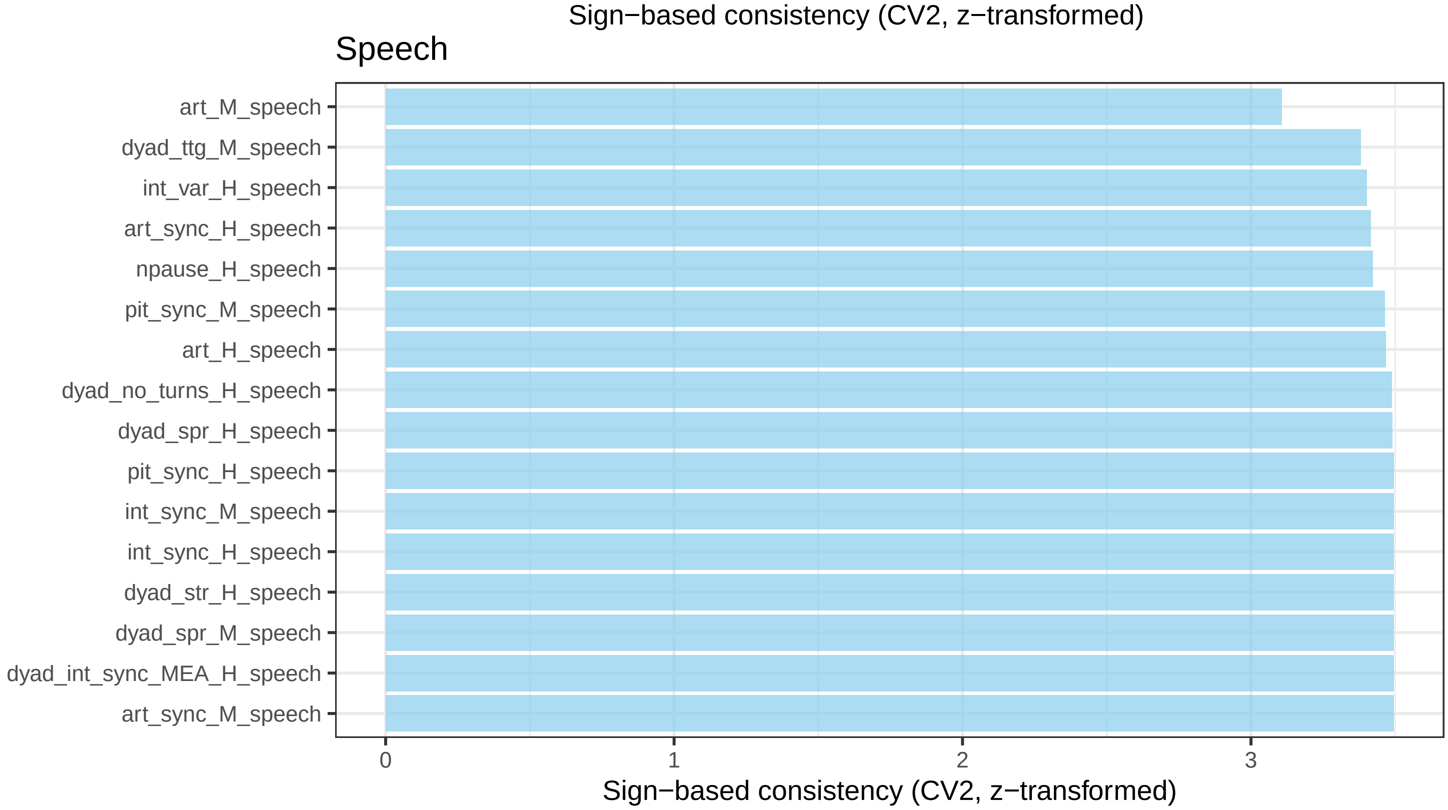

CROSSturn
self_md_H_AU07_r
other_md_H_AU07_r
self_mean_H_AU07_r
other_mean_H_AU07_r
self_mean_M_AU12_r
other_mean_M_AU12_r
self_mean_M_AU26_r
other_mean_M_AU26_r
self_sd_H_AU26_r
other_sd_H_AU26_r
self_sd_M_AU12_r
other_sd_M_AU12_r
self_sd_H_AU12_r
other_sd_H_AU12_r
self_max_M_AU12_r
other_max_M_AU12_r
0
1
2
3
Sign−based consistency (CV2, z−transformed)


FACEsync
md_M_AU26_r
max_M_AU14_r
sd_M_AU14_r
md_M_AU15_r
mean_H_AU26_r
min_H_AU26_r
max_H_AU07_r
md_H_AU45_r
sd_M_AU06_r
sd_M_AU26_r
max_M_AU06_r
max_H_AU26_r
mean_M_AU26_r
md_M_AU14_r
mean_M_AU14_r
sd_H_AU26_r
0
1
2
3
4
5
Sign−based consistency (CV2, z−transformed)

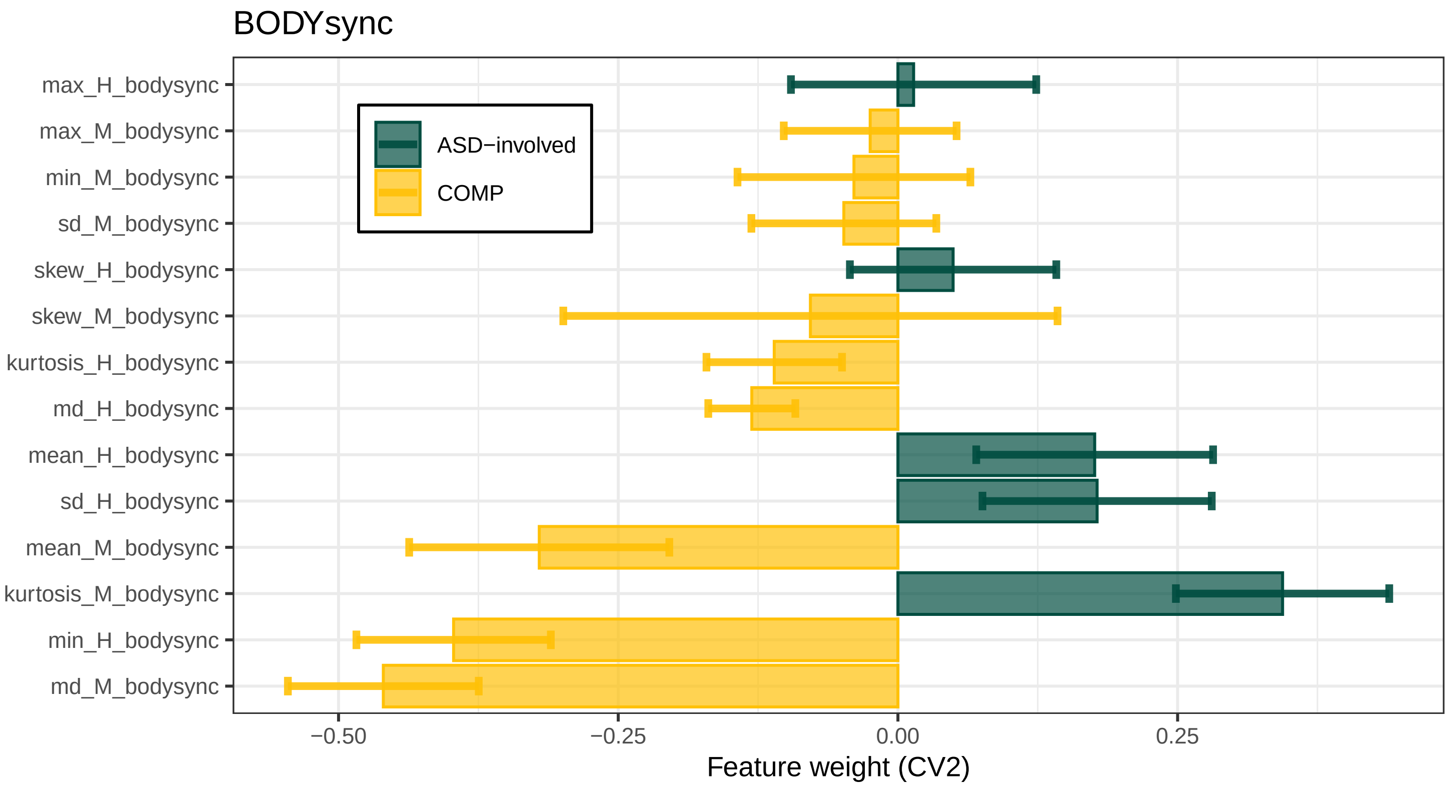

HEADsync
sd_M_pose_Rysync
md_H_pose_Rysync
mean_M_pose_Rysync
kurtosis_H_pose_Rysync
max_M_headsync
md_M_headsync
mean_M_headsync
mean_M_pose_Rxsync
sd_M_headsync
md_M_pose_Rysync
mean_H_pose_Rxsync
md_M_pose_Rxsync
min_H_pose_Rxsync
md_H_pose_Rxsync
kurtosis_H_pose_Rxsync
skew_H_pose_Rxsync
0
1
2
3
4
Sign−based consistency (CV2, z−transformed)

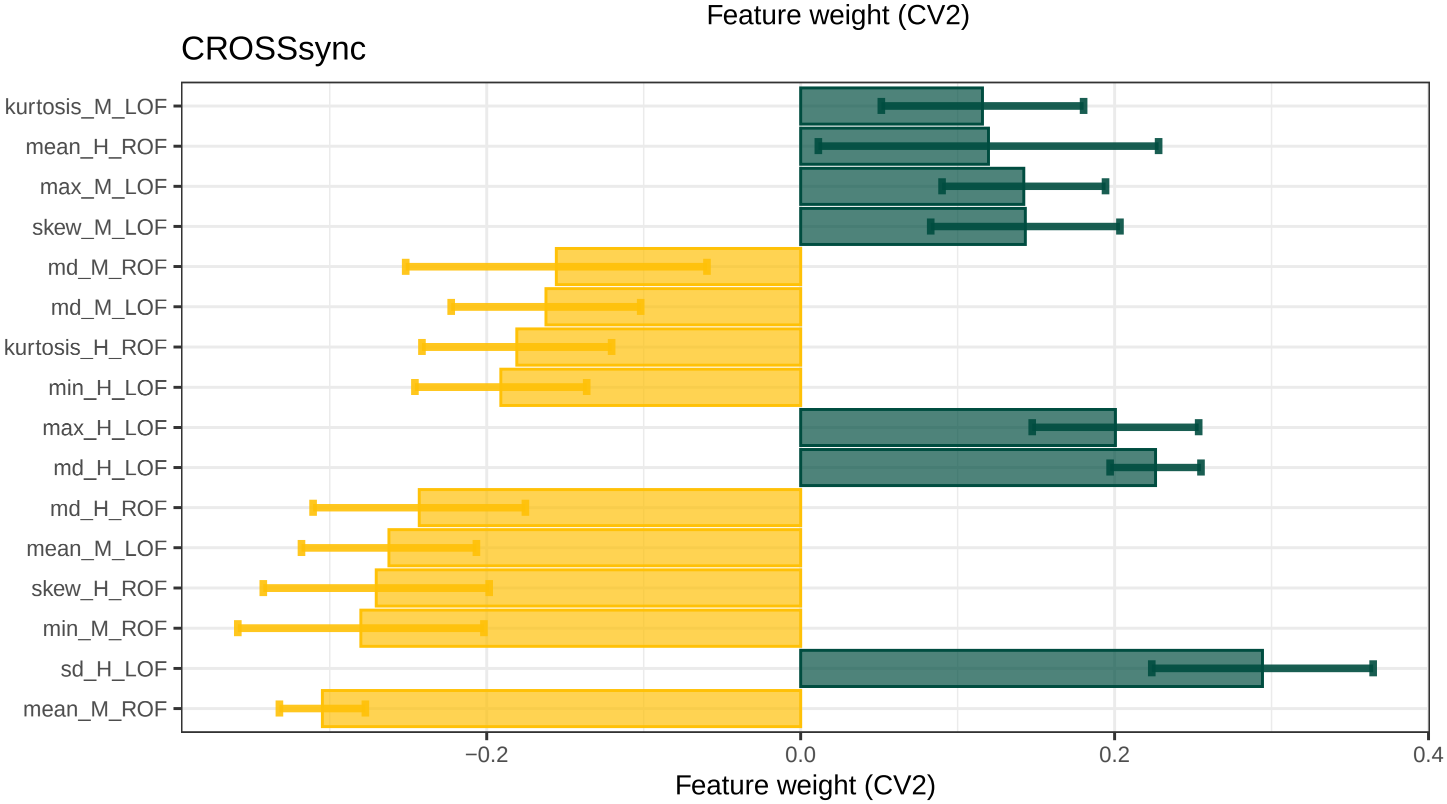

INTRAsync
kurtosis_M_intra
min_H_intra
max_M_intra
md_M_intra
md_H_intra
min_M_intra
sd_H_intra
kurtosis_H_intra
skew_M_intra
max_H_intra
mean_M_intra
sd_M_intra
skew_H_intra
mean_H_intra
0
1
2
3
Sign−based consistency (CV2, z−transformed)

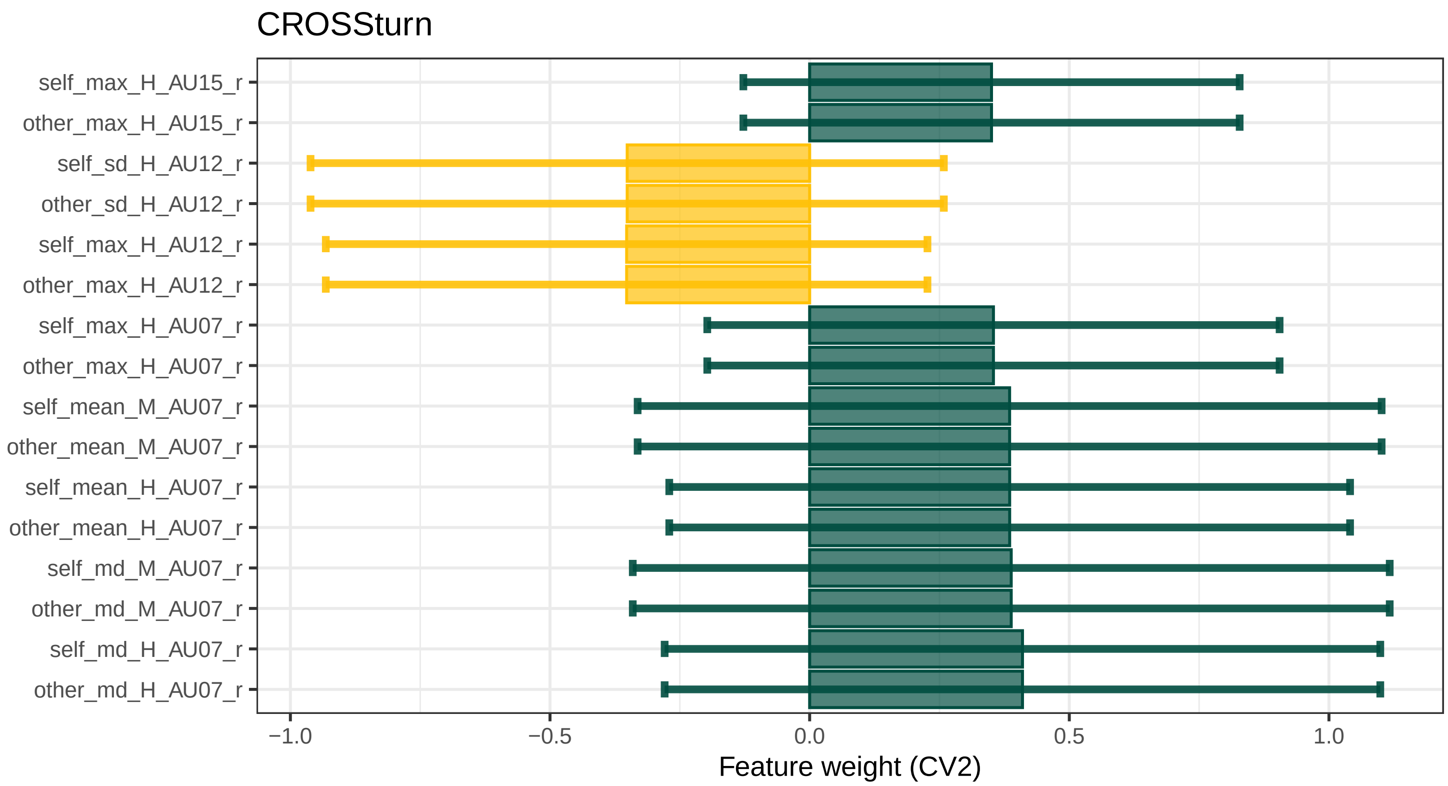

MovEx
H_head_total_movement
mean_intensity_M
M_body_total_movement
mean_intensity_H
H_body_total_movement
M_head_total_movement
0
5
10
15
20
Sign−based consistency (CV2, z−transformed)

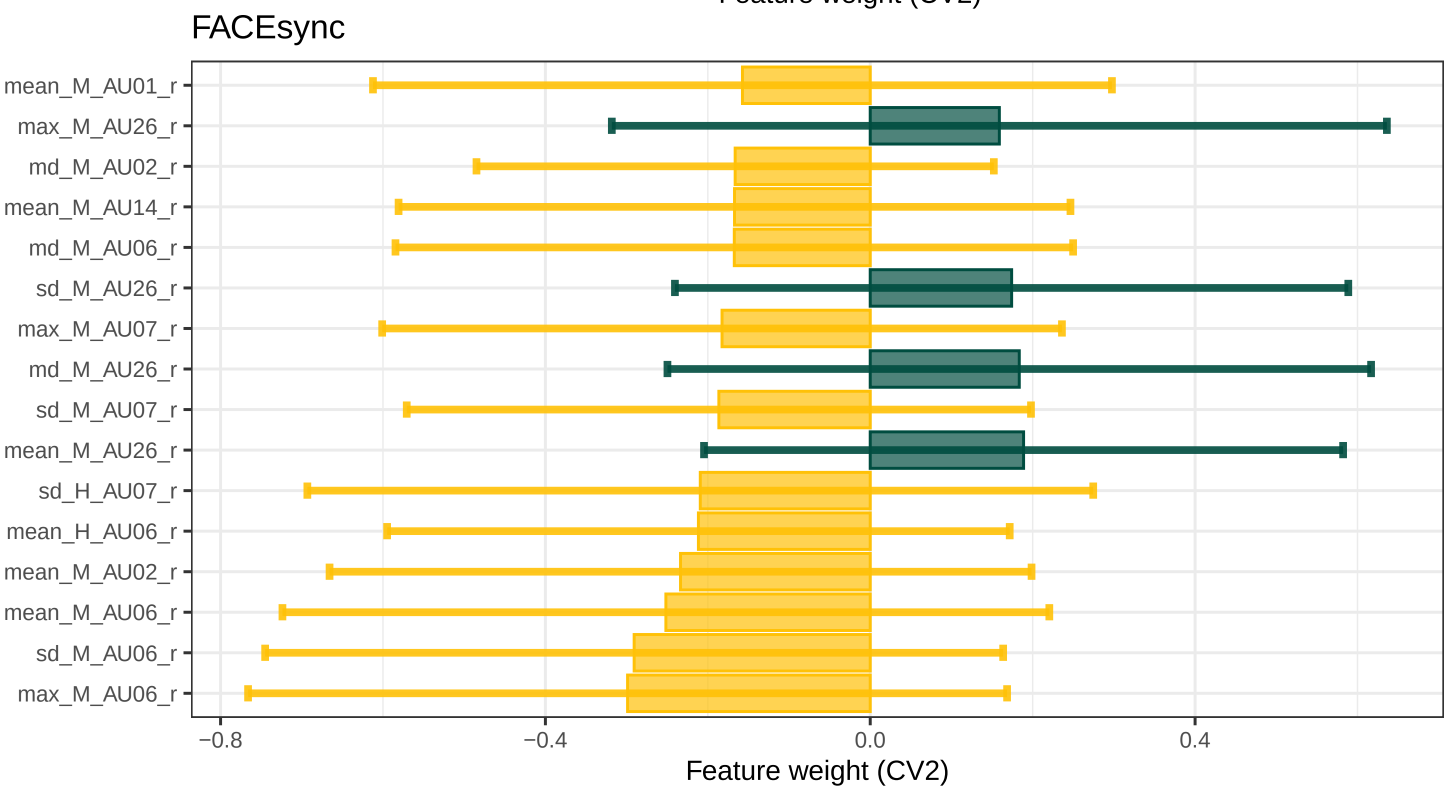

Speech
art_M_speech
dyad_ttg_M_speech
int_var_H_speech
art_sync_H_speech
npause_H_speech
pit_sync_M_speech
art_H_speech
dyad_no_turns_H_speech
dyad_spr_H_speech
pit_sync_H_speech
int_sync_M_speech
int_sync_H_speech
dyad_str_H_speech
dyad_spr_M_speech
dyad_int_sync_MEA_H_speech
art_sync_M_speech
0
1
2
3
Sign−based consistency (CV2, z−transformed)

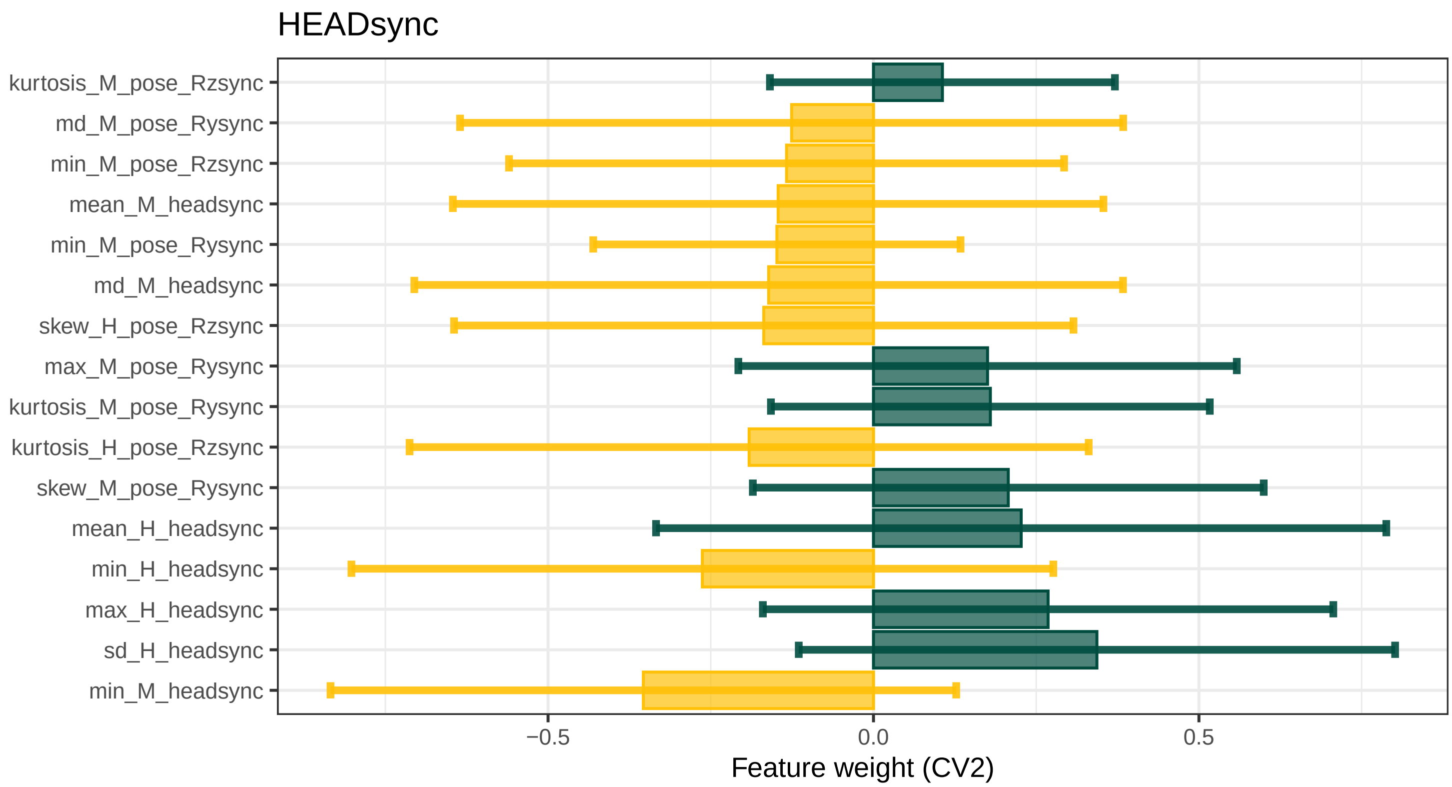
S6.2 ASD-involved versus COMP classifiers

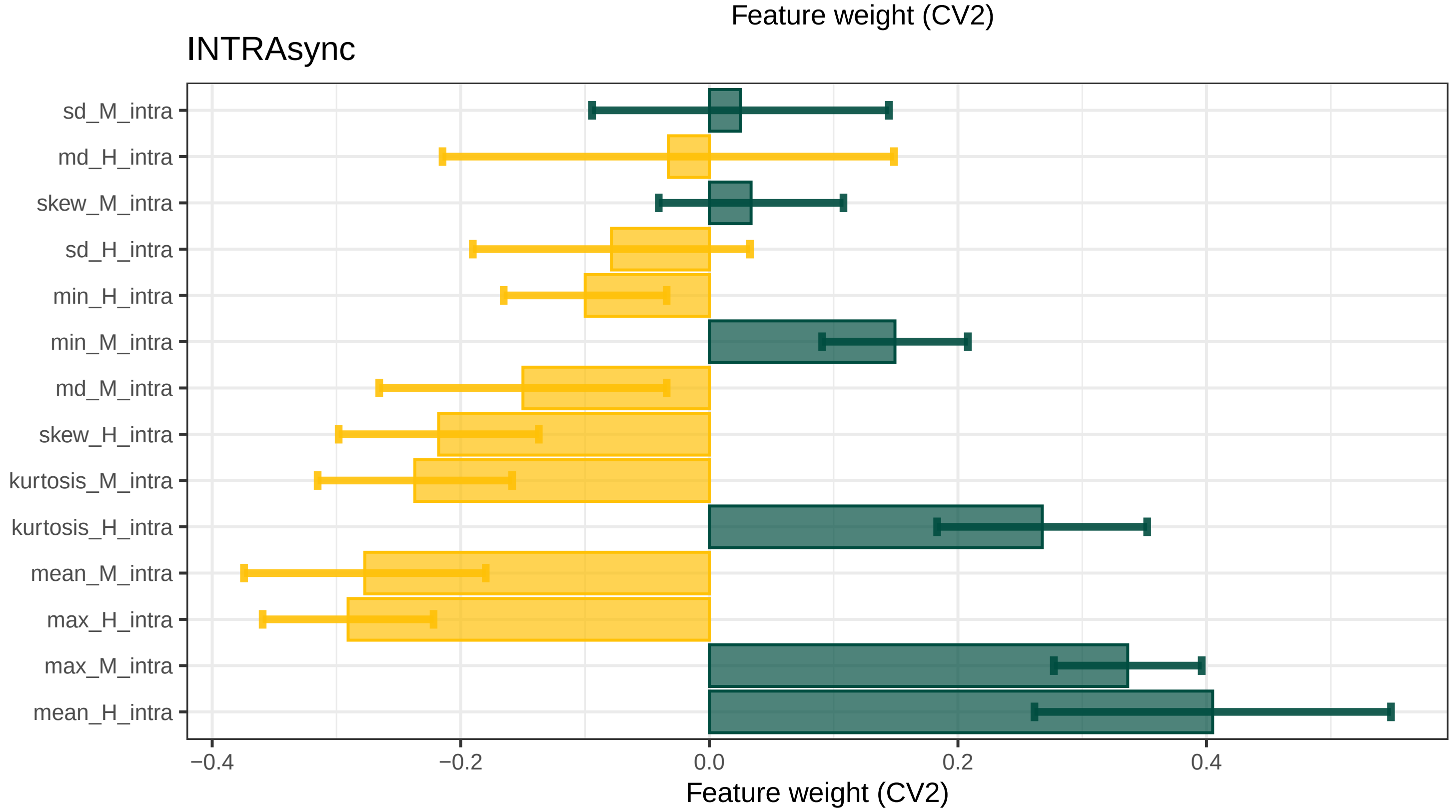


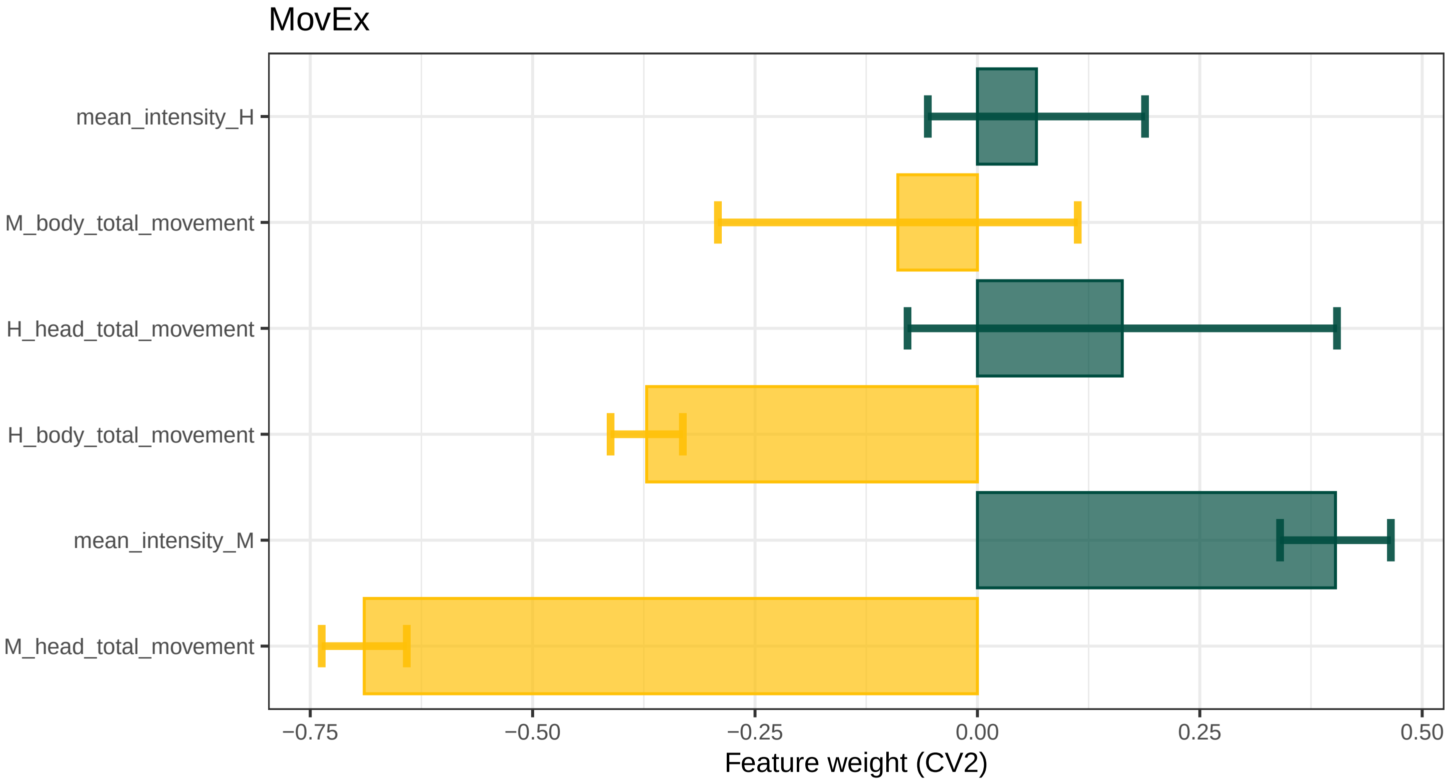



CROSSturn
self_max_H_AU15_r
other_max_H_AU15_r
self_sd_H_AU12_r
other_sd_H_AU12_r
self_max_H_AU12_r
other_max_H_AU12_r
self_max_H_AU07_r
other_max_H_AU07_r
self_mean_M_AU07_r
other_mean_M_AU07_r
self_mean_H_AU07_r
other_mean_H_AU07_r
self_md_M_AU07_r
other_md_M_AU07_r
self_md_H_AU07_r
other_md_H_AU07_r
−1.0
−0.5
0.0
0.5
1.0
Feature weight (CV2)

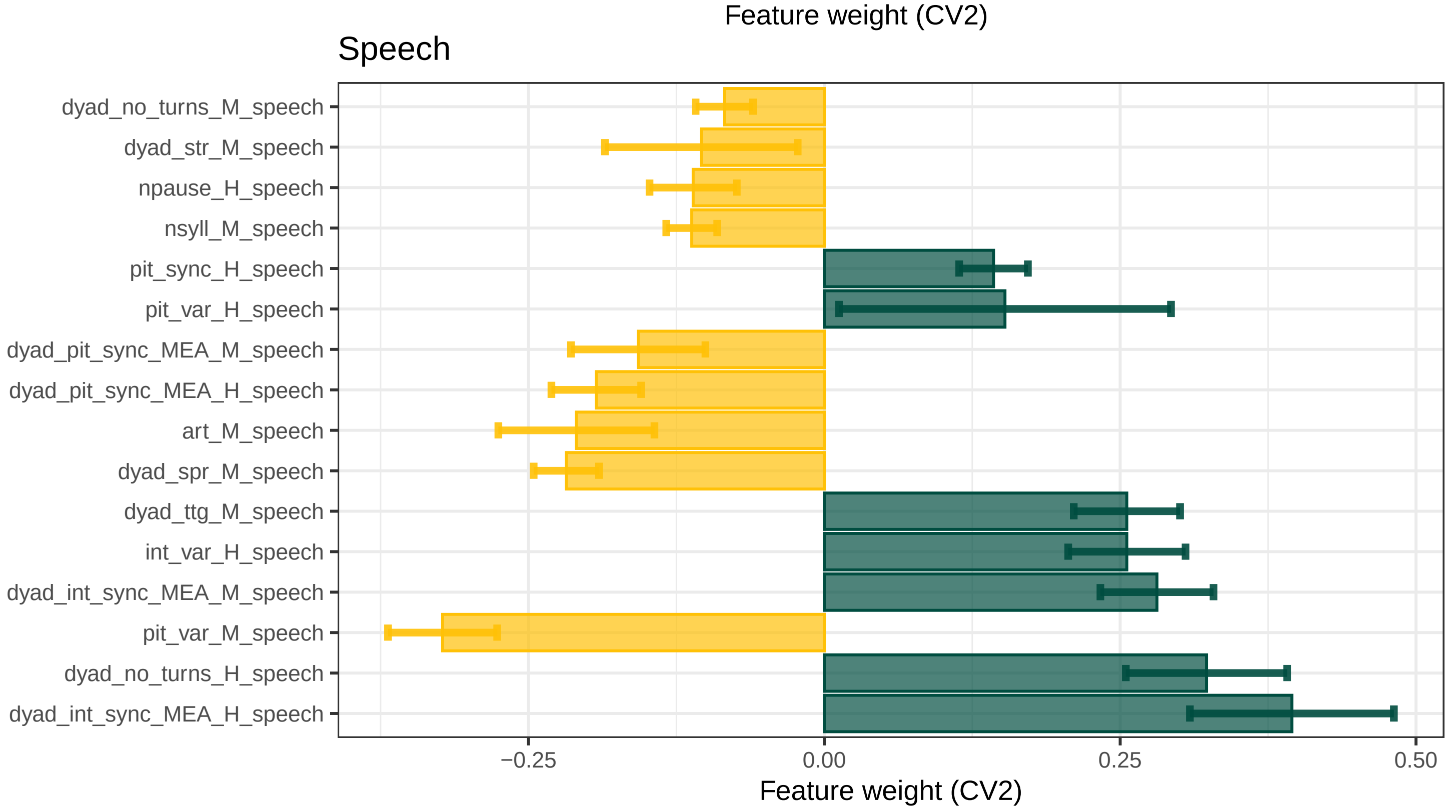

FACEsync
mean_M_AU01_r
max_M_AU26_r
md_M_AU02_r
mean_M_AU14_r
md_M_AU06_r
sd_M_AU26_r
max_M_AU07_r
md_M_AU26_r
sd_M_AU07_r
mean_M_AU26_r
sd_H_AU07_r
mean_H_AU06_r
mean_M_AU02_r
mean_M_AU06_r
sd_M_AU06_r
max_M_AU06_r
−0.8
−0.4
0.0
0.4
Feature weight (CV2)


HEADsync
kurtosis_M_pose_Rzsync
md_M_pose_Rysync
min_M_pose_Rzsync
mean_M_headsync
min_M_pose_Rysync
md_M_headsync
skew_H_pose_Rzsync
max_M_pose_Rysync
kurtosis_M_pose_Rysync
kurtosis_H_pose_Rzsync
skew_M_pose_Rysync
mean_H_headsync
min_H_headsync
max_H_headsync
sd_H_headsync
min_M_headsync
−0.5
0.0
0.5
Feature weight (CV2)

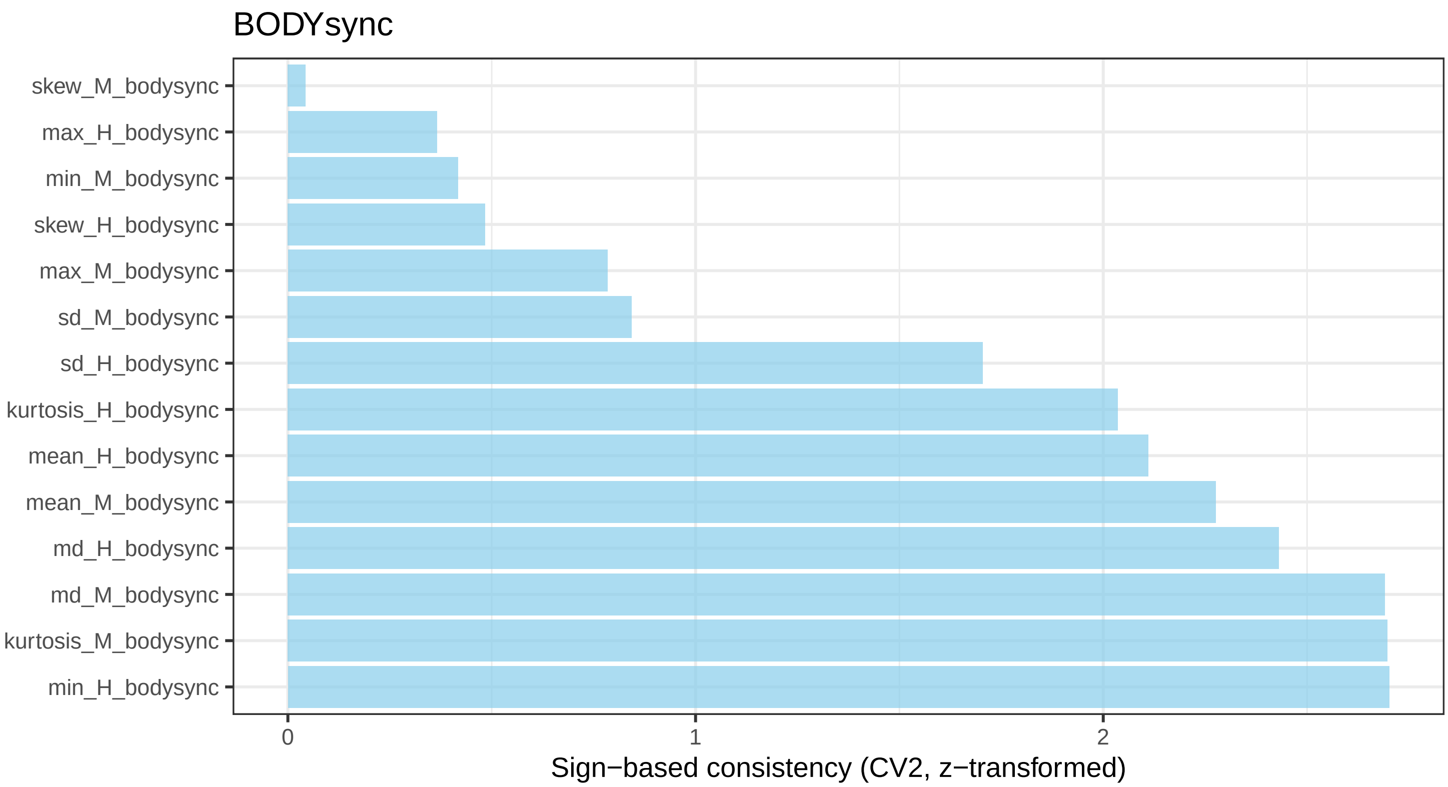

INTRAsync
sd_M_intra
md_H_intra
skew_M_intra
sd_H_intra
min_H_intra
min_M_intra
md_M_intra
skew_H_intra
kurtosis_M_intra
kurtosis_H_intra
mean_M_intra
max_H_intra
max_M_intra
mean_H_intra
−0.4
−0.2
0.0
0.2
0.4
Feature weight (CV2)

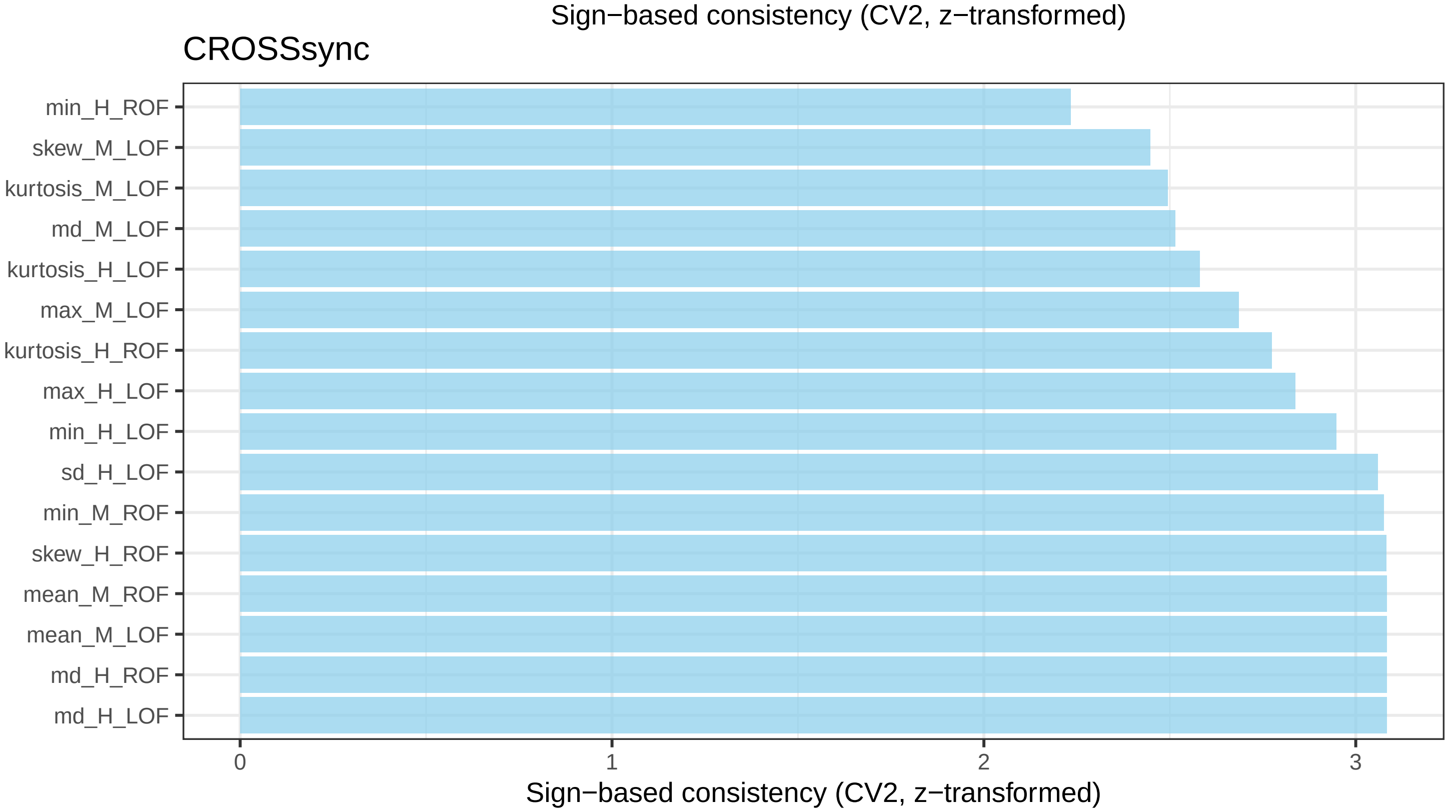


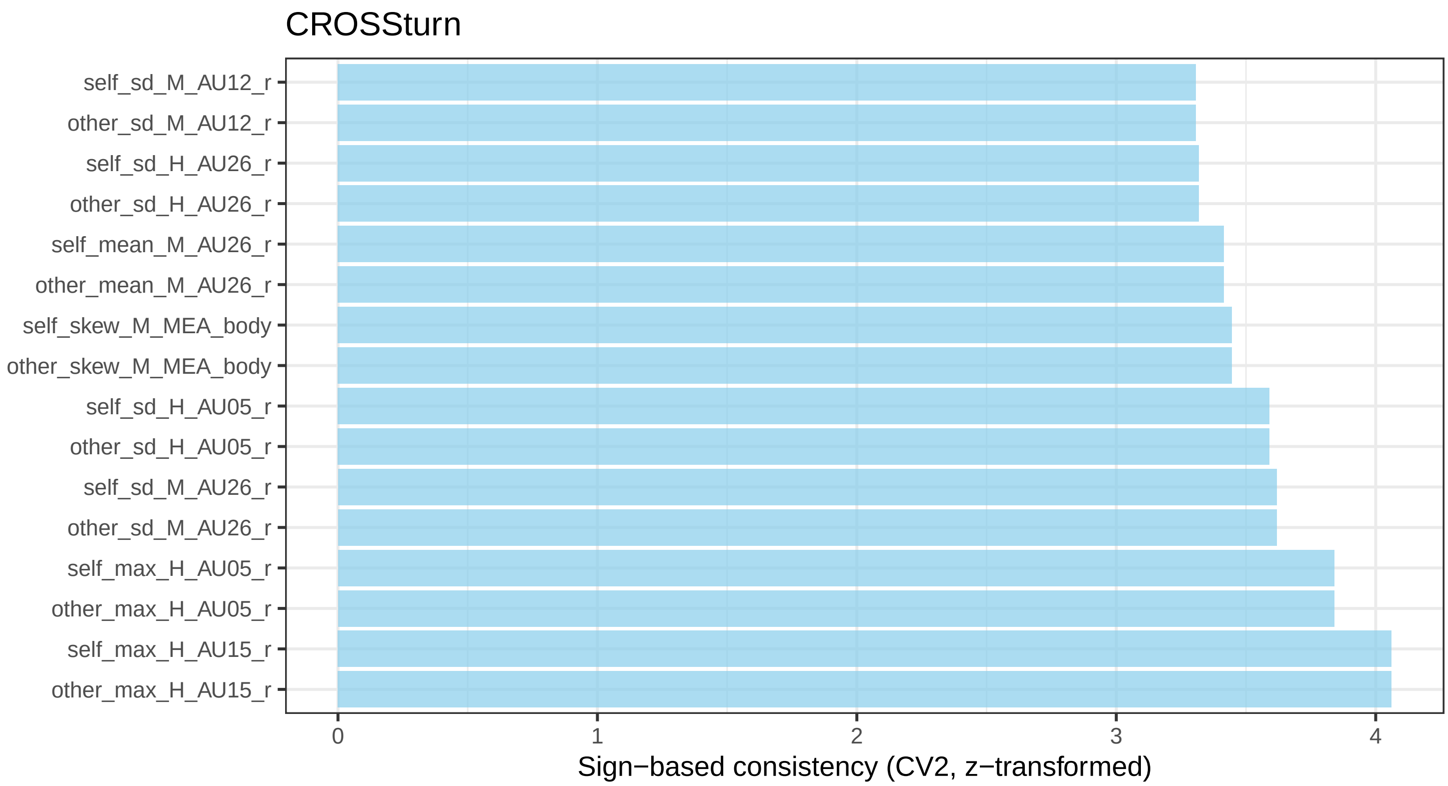
MovEx
mean_intensity_H
M_body_total_movement
H_head_total_movement
H_body_total_movement
mean_intensity_M
M_head_total_movement
−0.75
−0.50
−0.25
0.00
0.25
0.50
Feature weight (CV2)

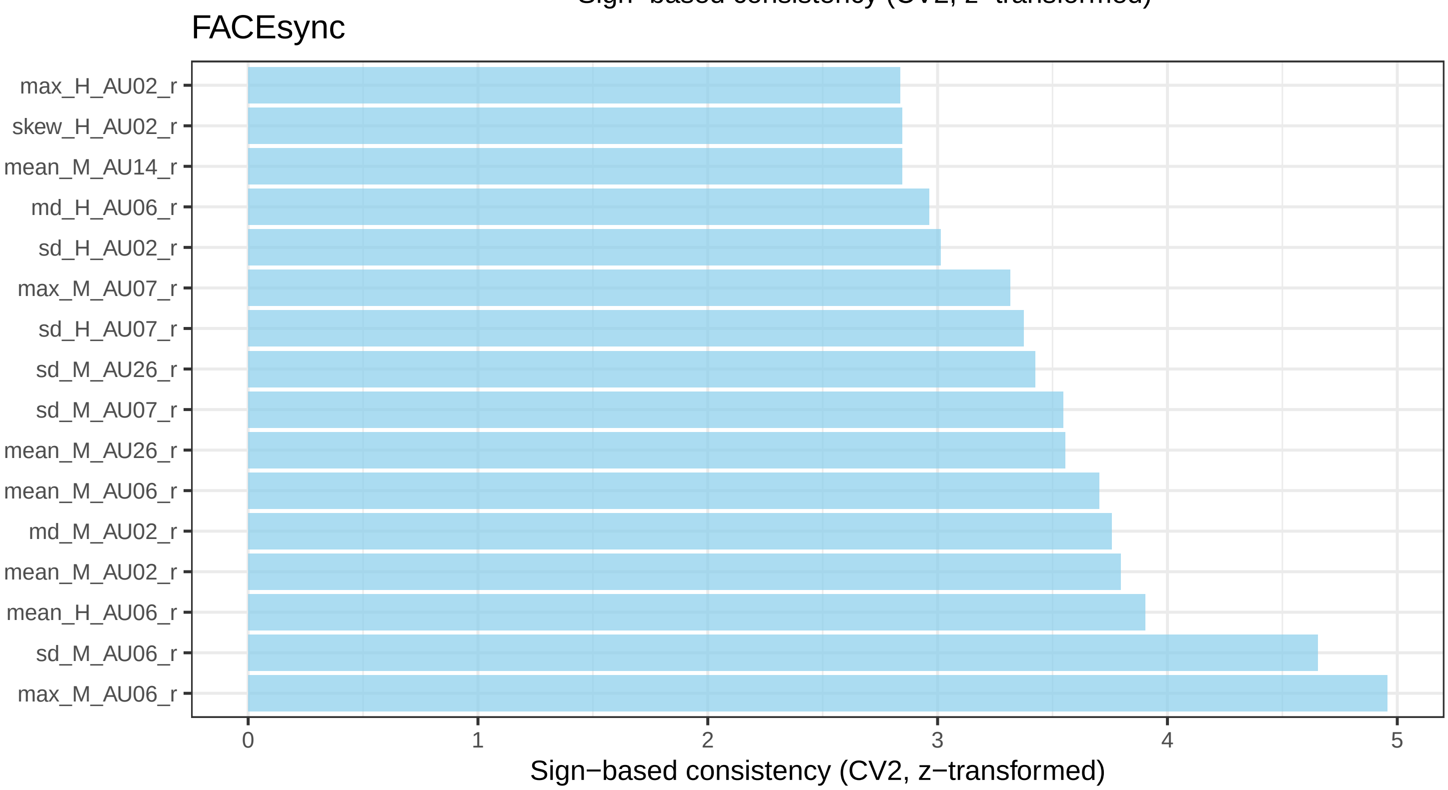


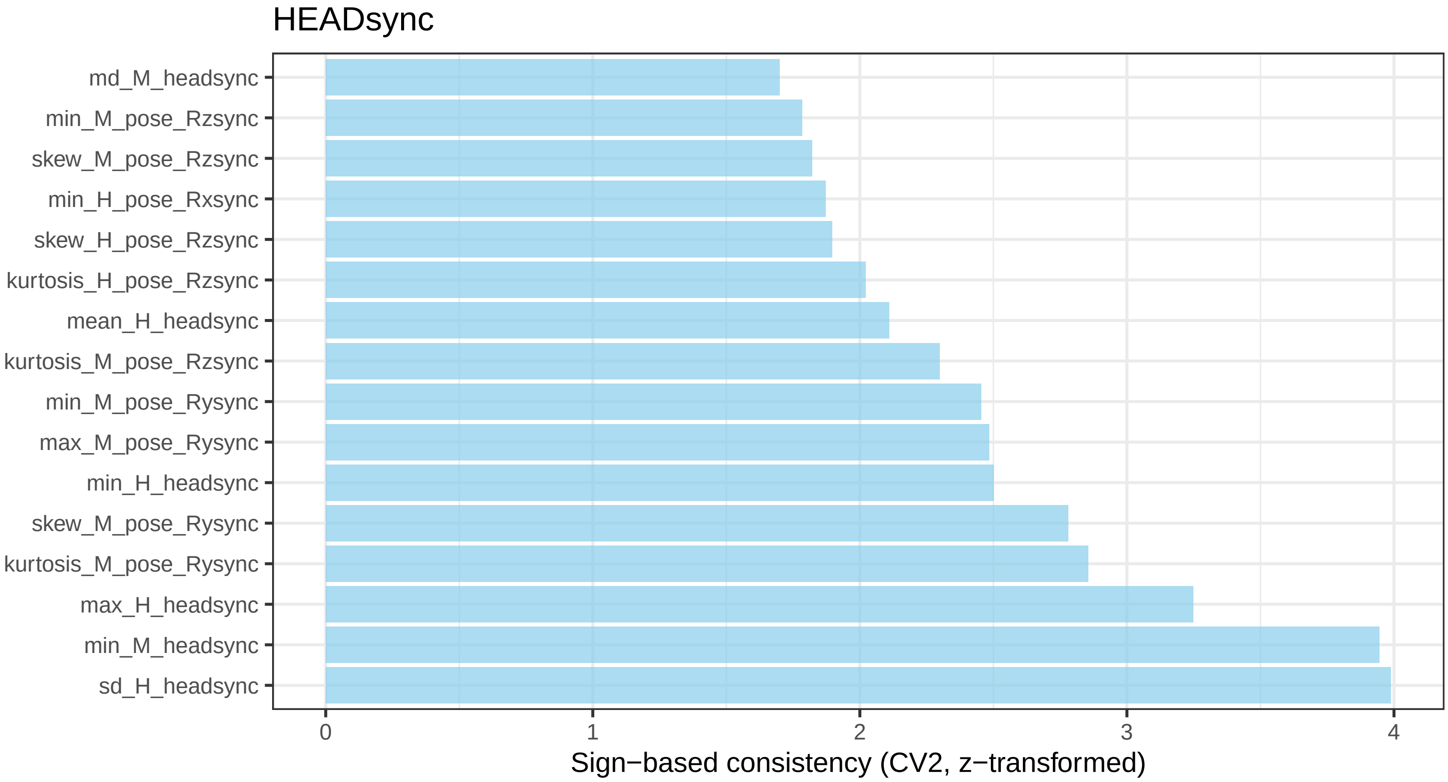
Speech
dyad_no_turns_M_speech
dyad_str_M_speech
npause_H_speech
nsyll_M_speech
pit_sync_H_speech
pit_var_H_speech
dyad_pit_sync_MEA_M_speech
dyad_pit_sync_MEA_H_speech
art_M_speech
dyad_spr_M_speech
dyad_ttg_M_speech
int_var_H_speech
dyad_int_sync_MEA_M_speech
pit_var_M_speech
dyad_no_turns_H_speech
dyad_int_sync_MEA_H_speech
−0.25
0.00
0.25
0.50
Feature weight (CV2)

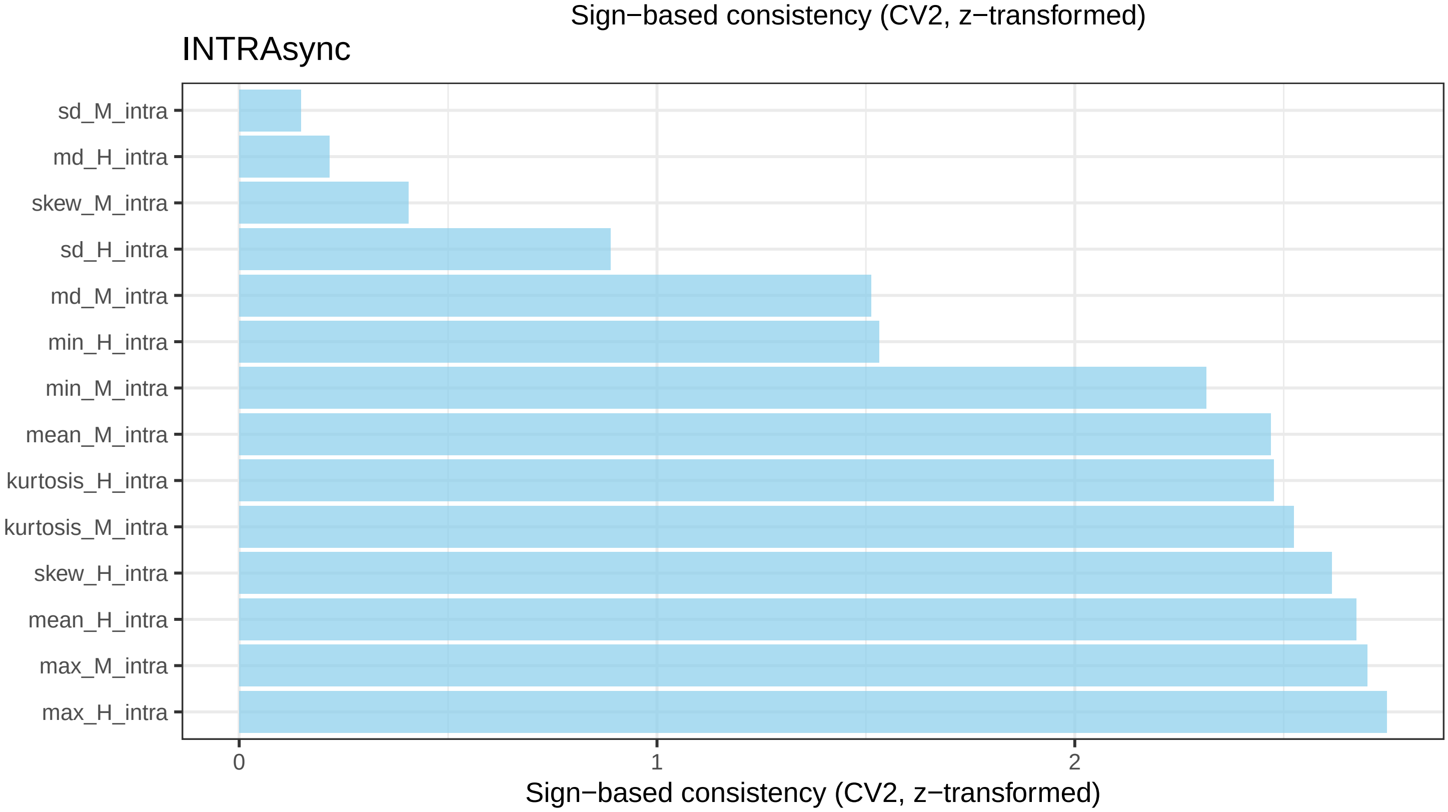

BODYsync
skew_M_bodysync
max_H_bodysync
min_M_bodysync
skew_H_bodysync
max_M_bodysync
sd_M_bodysync
sd_H_bodysync
kurtosis_H_bodysync
mean_H_bodysync
mean_M_bodysync
md_H_bodysync
md_M_bodysync
kurtosis_M_bodysync
min_H_bodysync
0
1
2
Sign−based consistency (CV2, z−transformed)

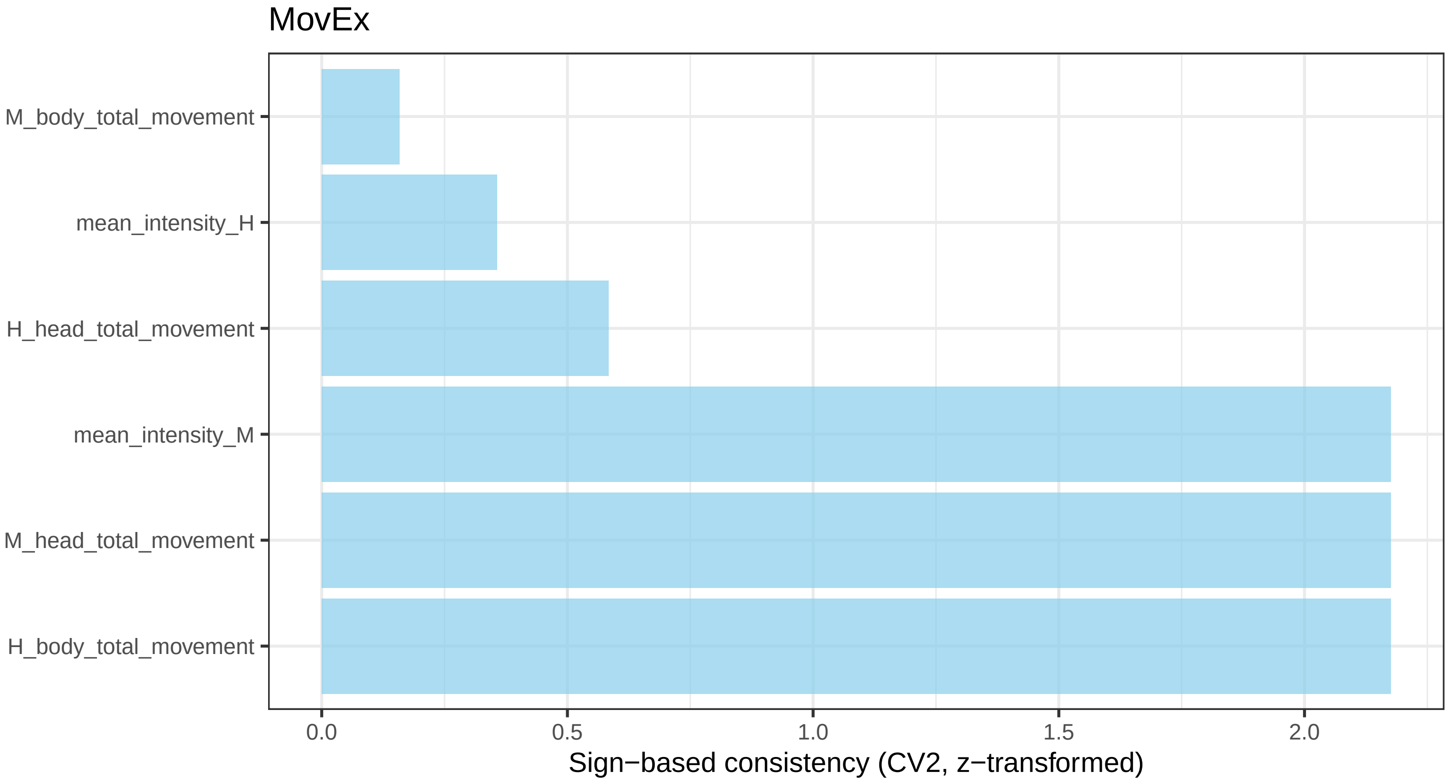

CROSSsync
min_H_ROF
skew_M_LOF
kurtosis_M_LOF
md_M_LOF
kurtosis_H_LOF
max_M_LOF
kurtosis_H_ROF
max_H_LOF
min_H_LOF
sd_H_LOF
min_M_ROF
skew_H_ROF
mean_M_ROF
mean_M_LOF
md_H_ROF
md_H_LOF
0
1
2
3
Sign−based consistency (CV2, z−transformed)

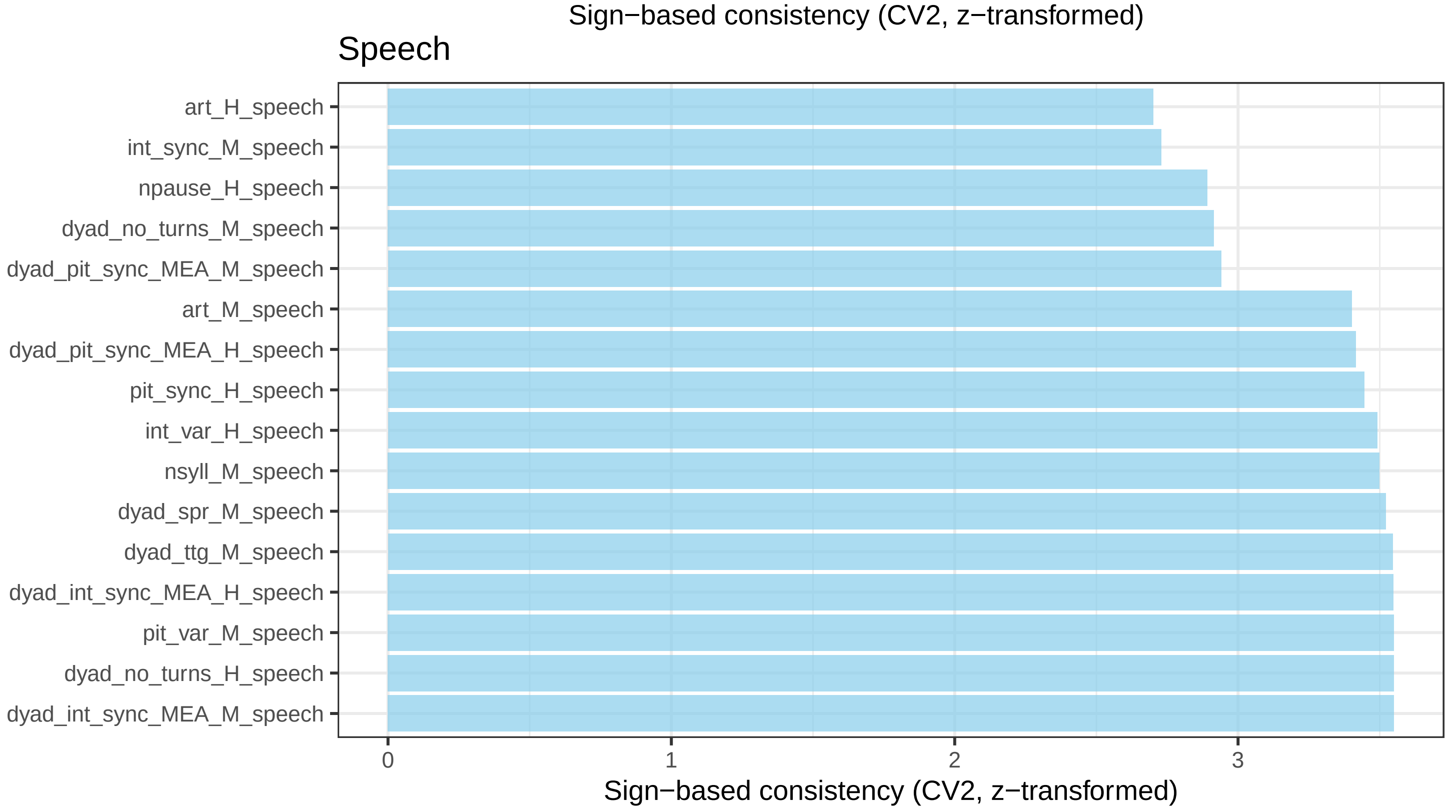

CROSSturn
self_sd_M_AU12_r
other_sd_M_AU12_r
self_sd_H_AU26_r
other_sd_H_AU26_r
self_mean_M_AU26_r
other_mean_M_AU26_r
self_skew_M_MEA_body
other_skew_M_MEA_body
self_sd_H_AU05_r
other_sd_H_AU05_r
self_sd_M_AU26_r
other_sd_M_AU26_r
self_max_H_AU05_r
other_max_H_AU05_r
self_max_H_AU15_r
other_max_H_AU15_r
0
1
2
3
4
Sign−based consistency (CV2, z−transformed)


FACEsync
max_H_AU02_r
skew_H_AU02_r
mean_M_AU14_r
md_H_AU06_r
sd_H_AU02_r
max_M_AU07_r
sd_H_AU07_r
sd_M_AU26_r
sd_M_AU07_r
mean_M_AU26_r
mean_M_AU06_r
md_M_AU02_r
mean_M_AU02_r
mean_H_AU06_r
sd_M_AU06_r
max_M_AU06_r
0
1
2
3
4
5
Sign−based consistency (CV2, z−transformed)

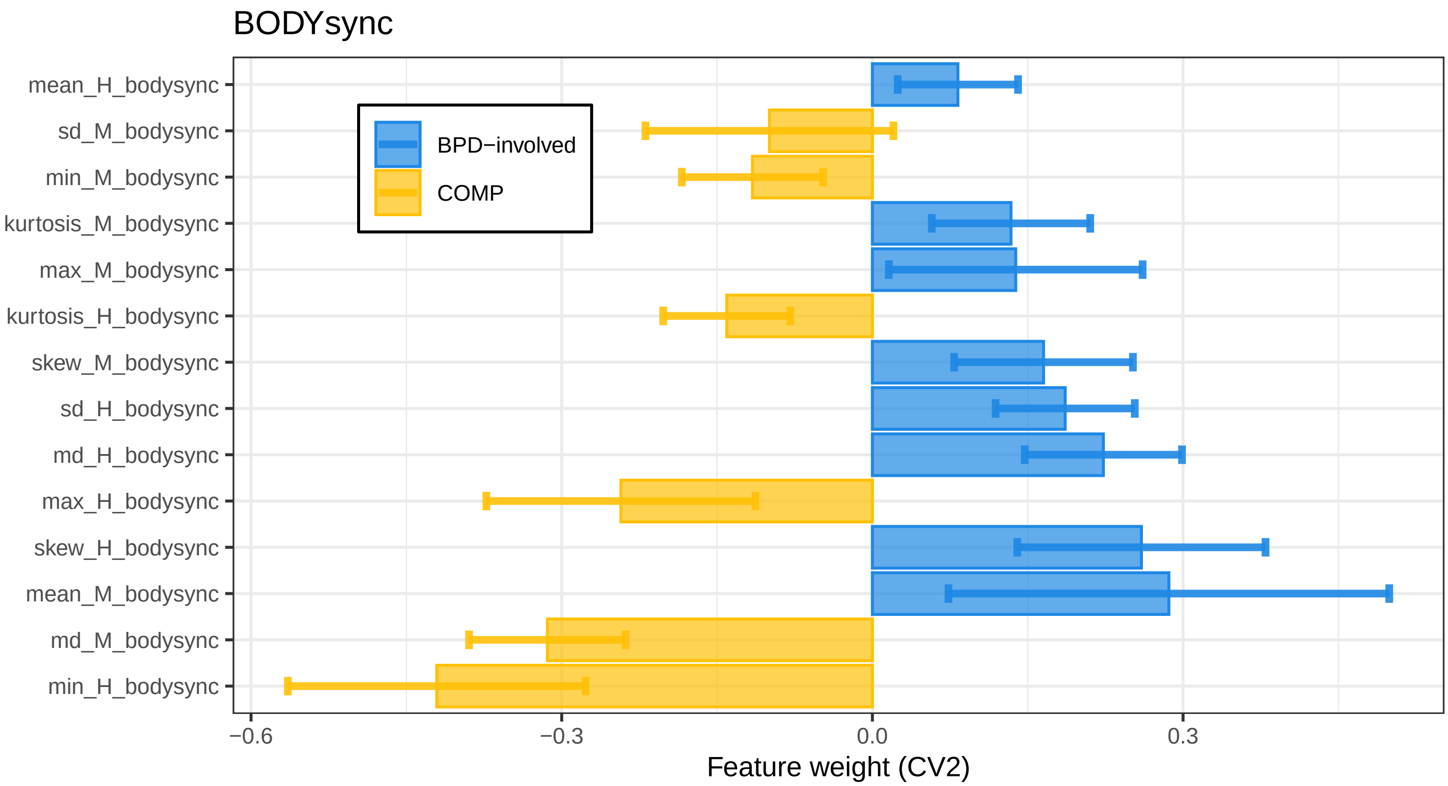

HEADsync
md_M_headsync
min_M_pose_Rzsync
skew_M_pose_Rzsync
min_H_pose_Rxsync
skew_H_pose_Rzsync
kurtosis_H_pose_Rzsync
mean_H_headsync
kurtosis_M_pose_Rzsync
min_M_pose_Rysync
max_M_pose_Rysync
min_H_headsync
skew_M_pose_Rysync
kurtosis_M_pose_Rysync
max_H_headsync
min_M_headsync
sd_H_headsync
0
1
2
3
4
Sign−based consistency (CV2, z−transformed)

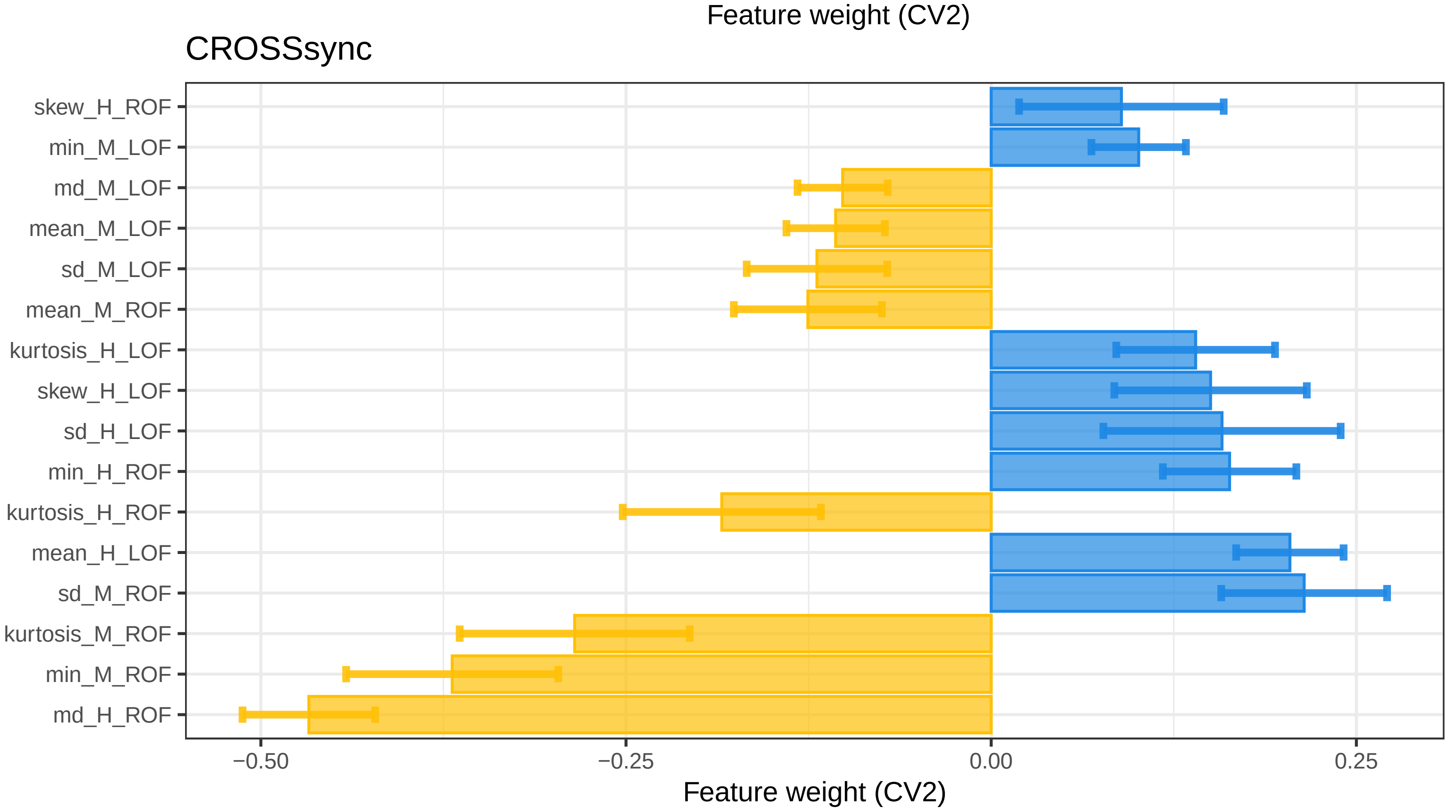

INTRAsync
sd_M_intra
md_H_intra
skew_M_intra
sd_H_intra
md_M_intra
min_H_intra
min_M_intra
mean_M_intra
kurtosis_H_intra
kurtosis_M_intra
skew_H_intra
mean_H_intra
max_M_intra
max_H_intra
0
1
2
Sign−based consistency (CV2, z−transformed)

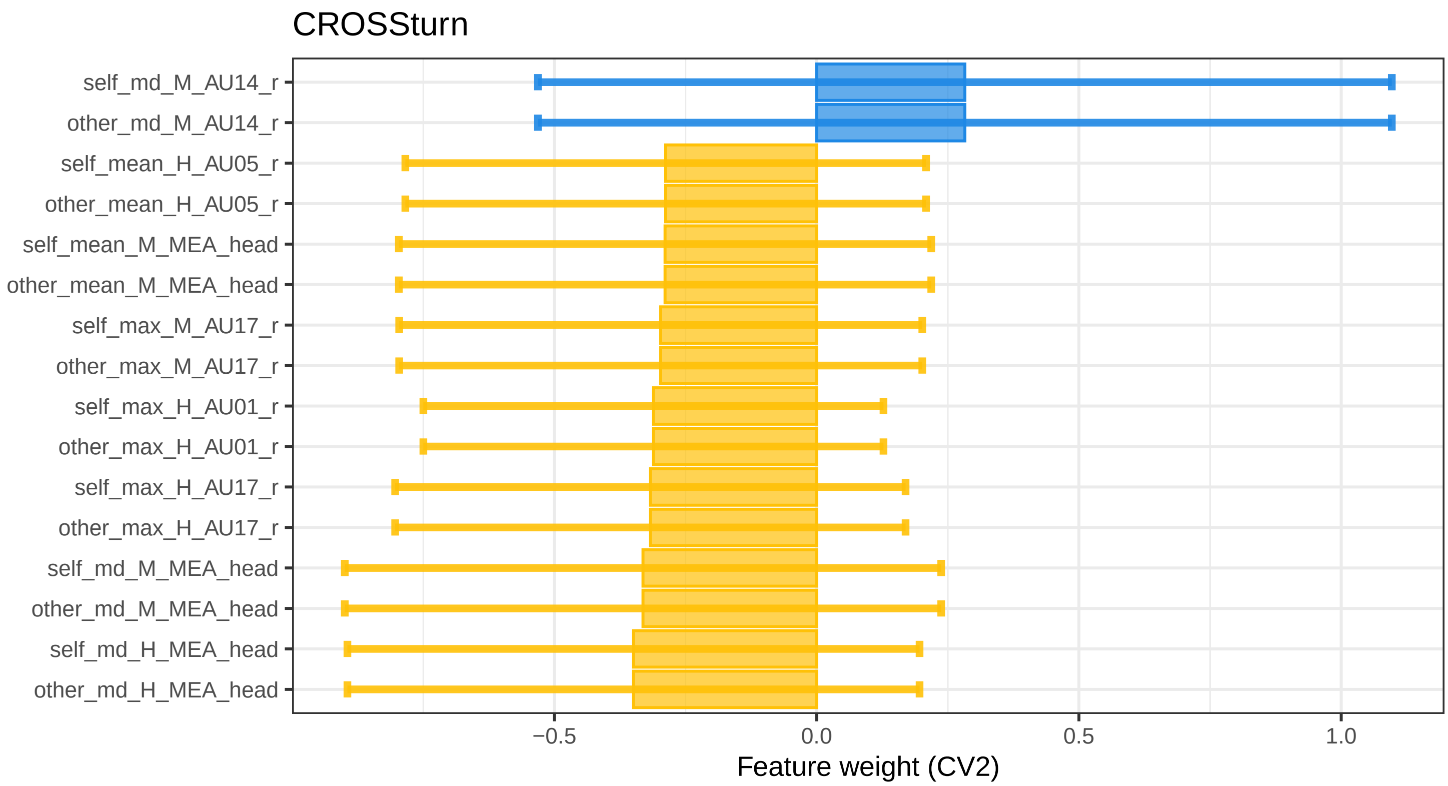

MovEx
M_body_total_movement
mean_intensity_H
H_head_total_movement
mean_intensity_M
M_head_total_movement
H_body_total_movement
0.0
0.5
1.0
1.5
2.0
Sign−based consistency (CV2, z−transformed)

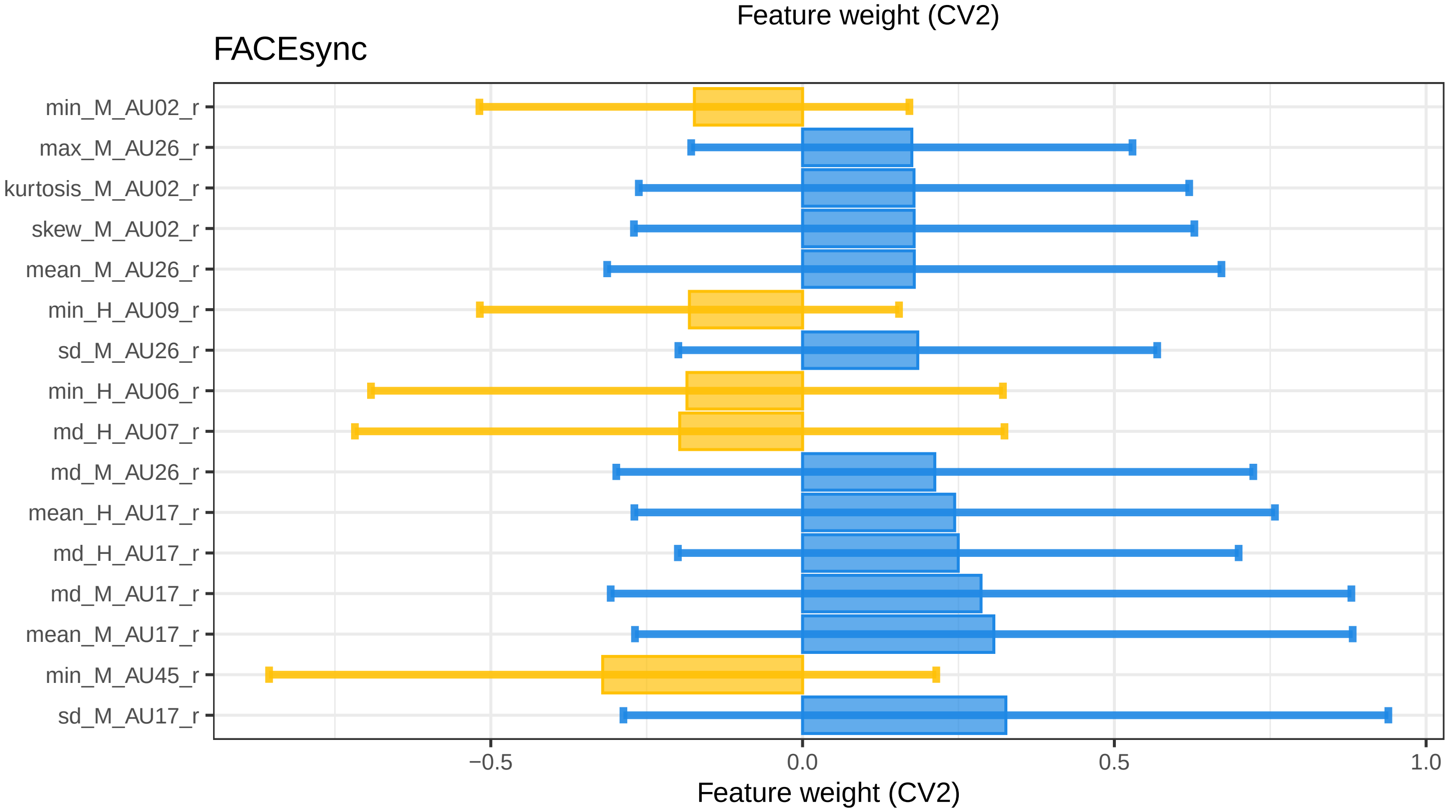

Speech
art_H_speech
int_sync_M_speech
npause_H_speech
dyad_no_turns_M_speech
dyad_pit_sync_MEA_M_speech
art_M_speech
dyad_pit_sync_MEA_H_speech
pit_sync_H_speech
int_var_H_speech
nsyll_M_speech
dyad_spr_M_speech
dyad_ttg_M_speech
dyad_int_sync_MEA_H_speech
pit_var_M_speech
dyad_no_turns_H_speech
dyad_int_sync_MEA_M_speech
0
1
2
3
Sign−based consistency (CV2, z−transformed)

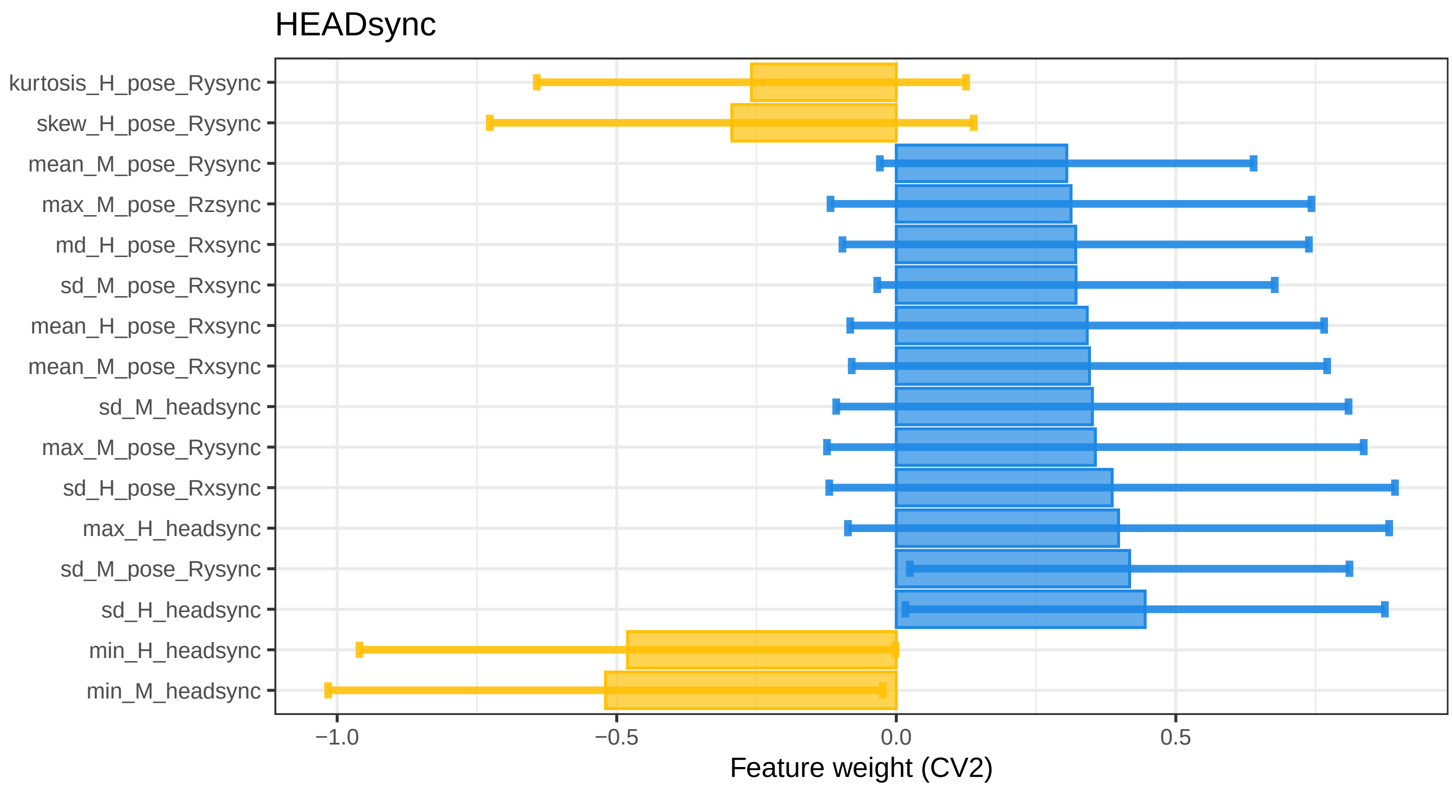
S6.3 ASD-involved versus COMP classifiers

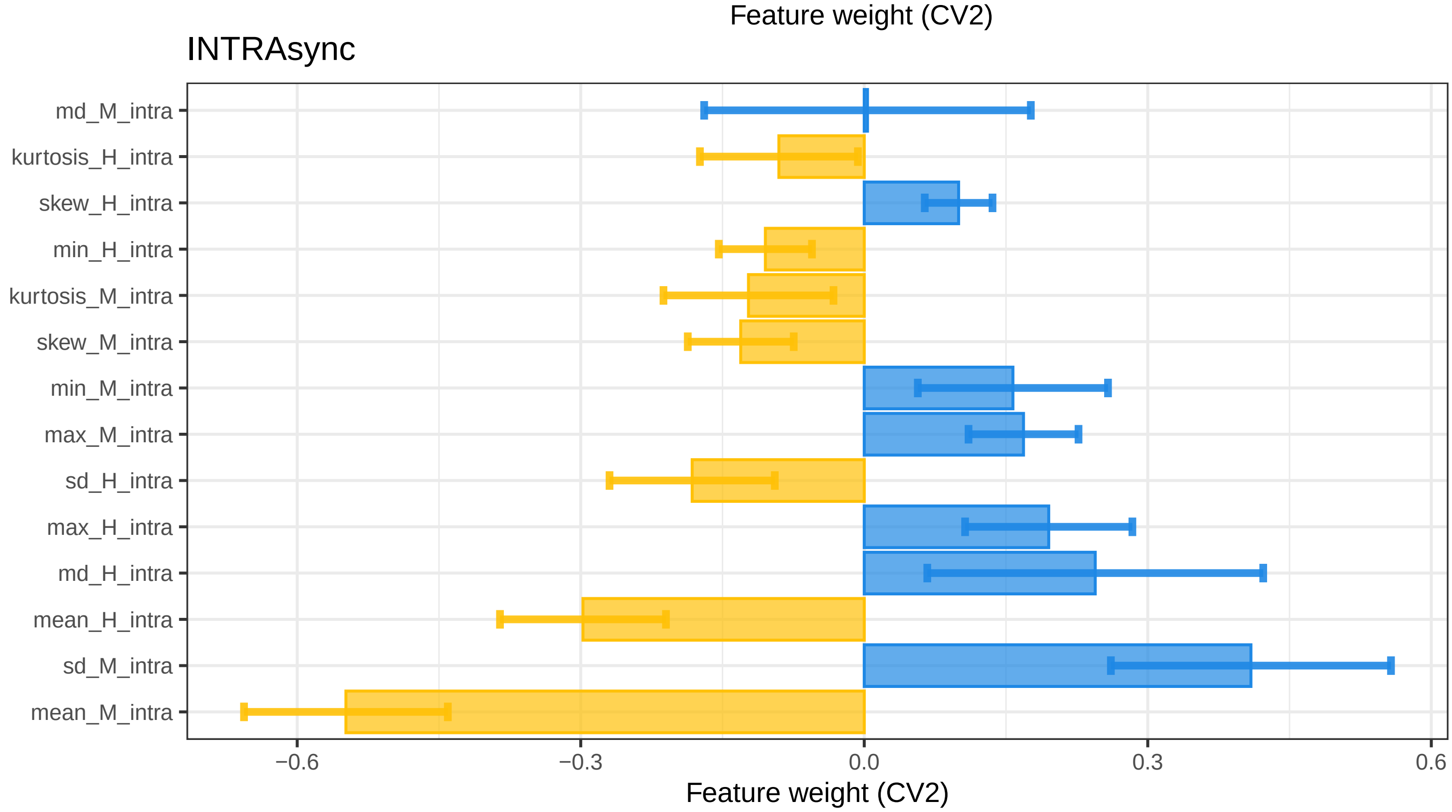


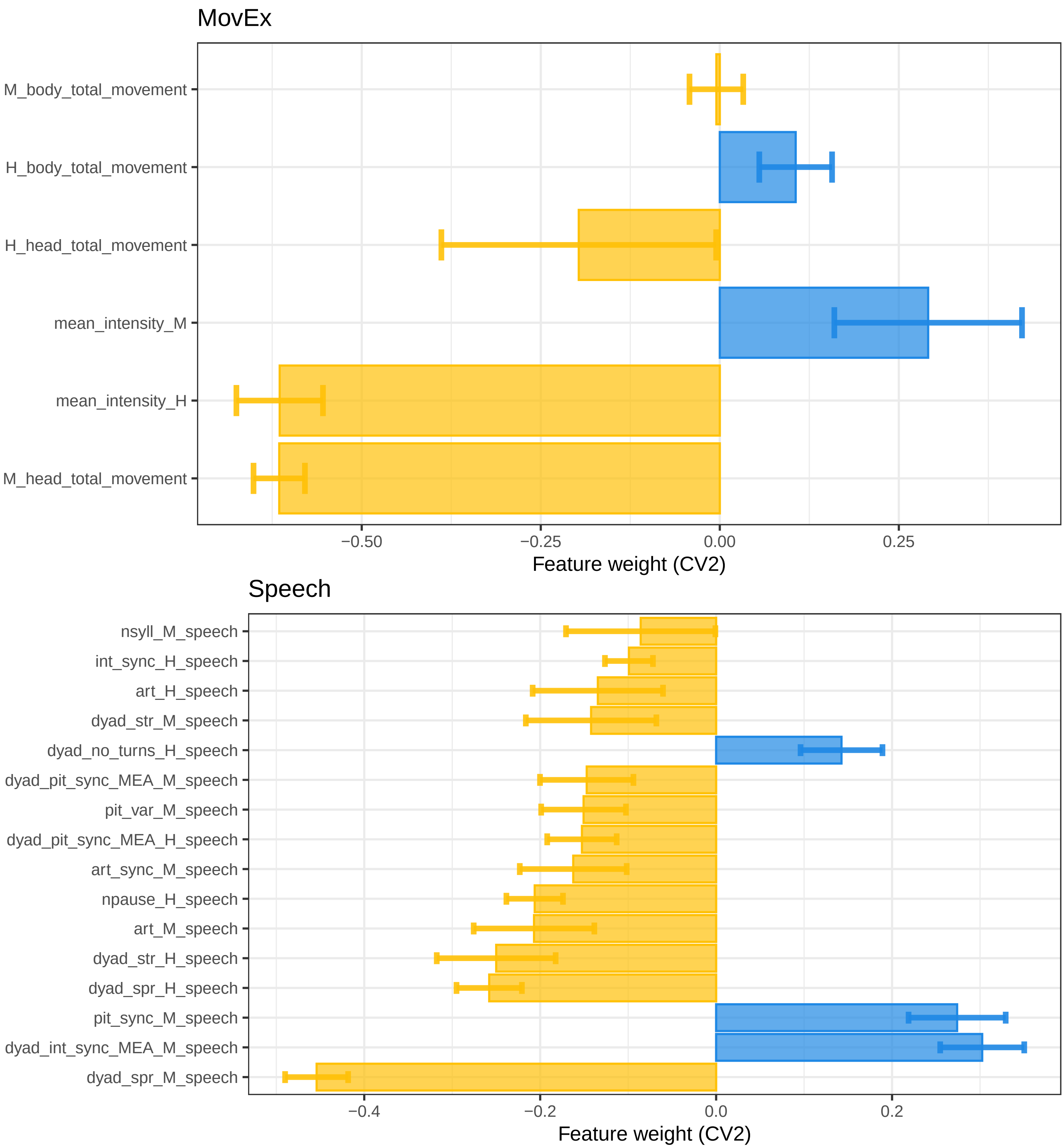


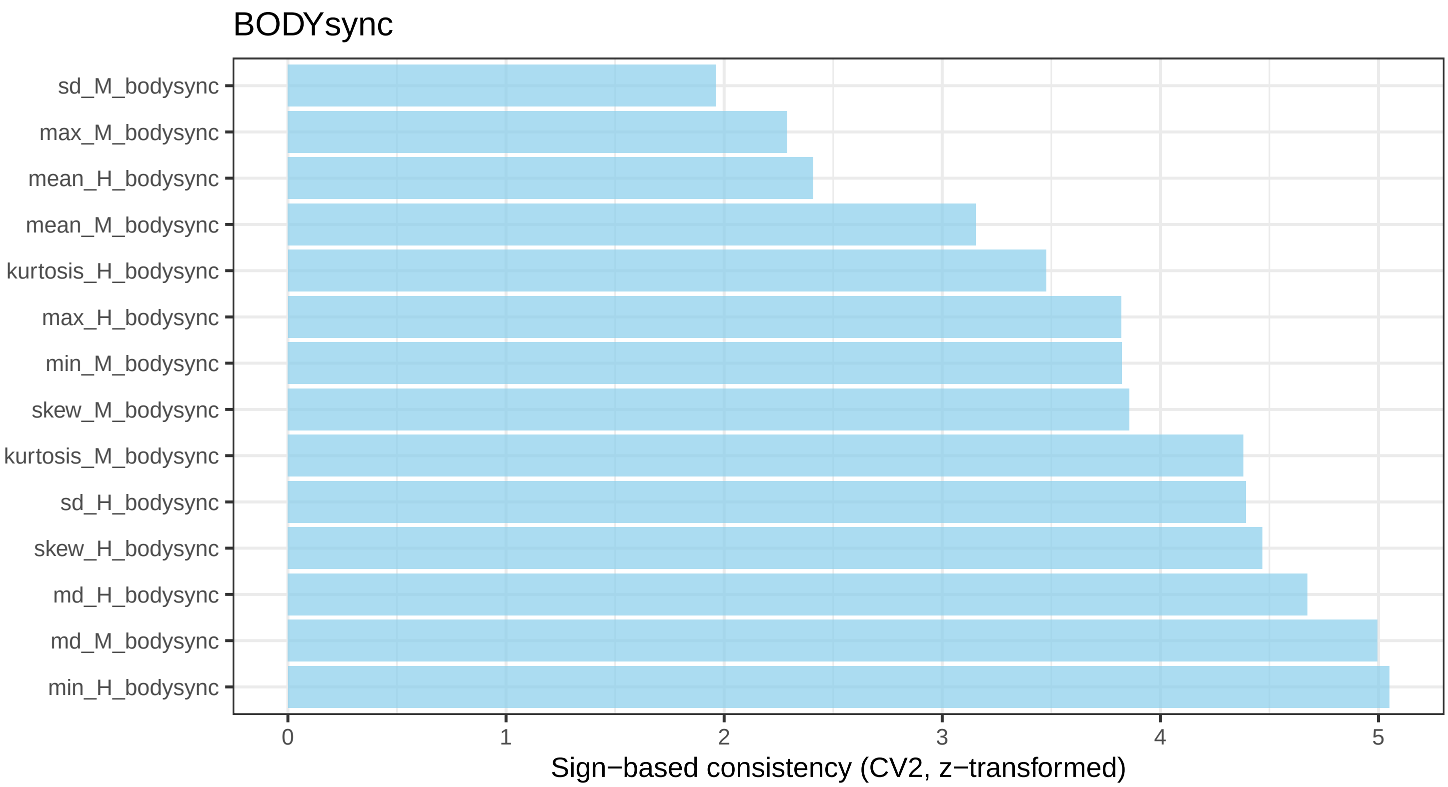

CROSSturn
self_md_M_AU14_r
other_md_M_AU14_r
self_mean_H_AU05_r
other_mean_H_AU05_r
self_mean_M_MEA_head
other_mean_M_MEA_head
self_max_M_AU17_r
other_max_M_AU17_r
self_max_H_AU01_r
other_max_H_AU01_r
self_max_H_AU17_r
other_max_H_AU17_r
self_md_M_MEA_head
other_md_M_MEA_head
self_md_H_MEA_head
other_md_H_MEA_head
−0.5
0.0
0.5
1.0
Feature weight (CV2)

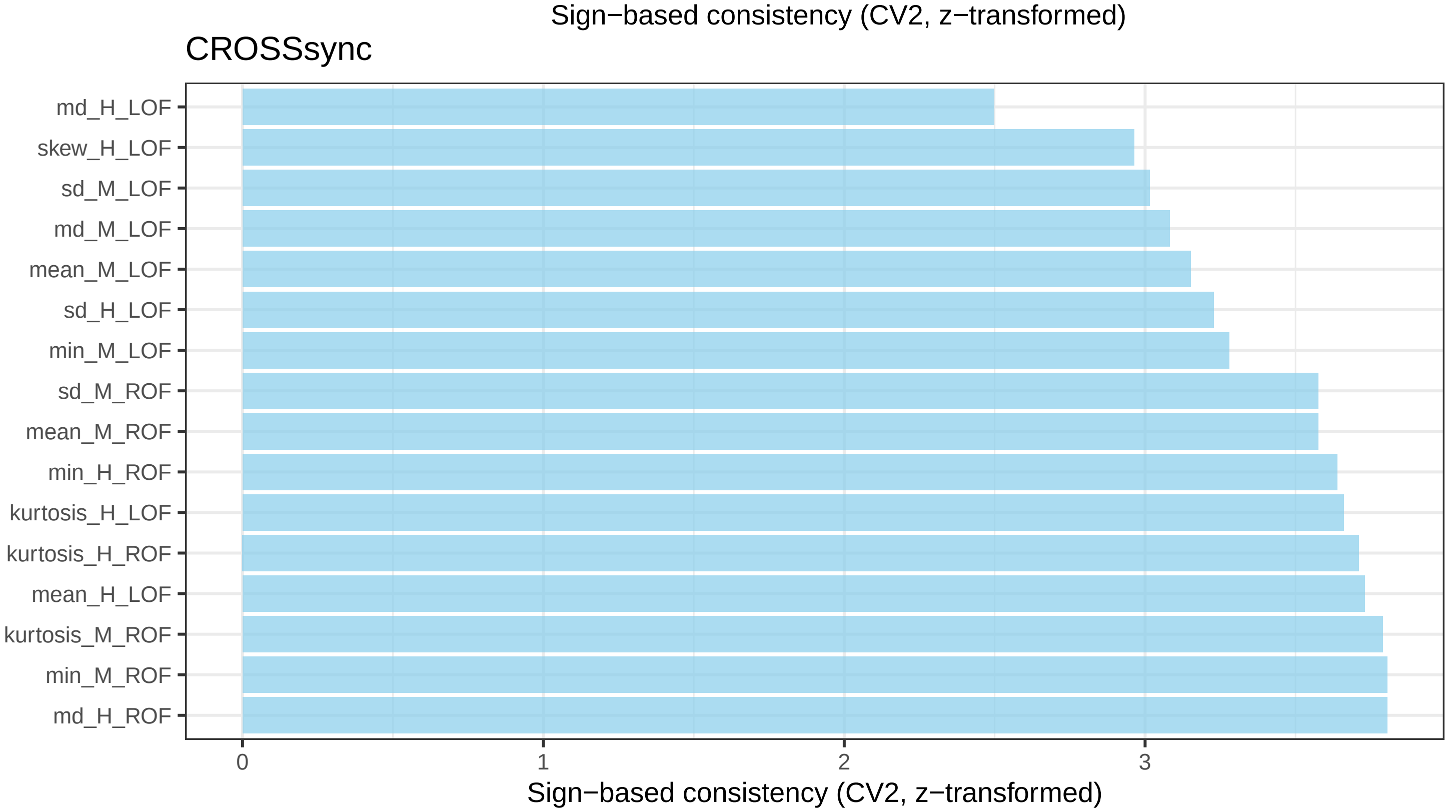

FACEsync
min_M_AU02_r
max_M_AU26_r
kurtosis_M_AU02_r
skew_M_AU02_r
mean_M_AU26_r
min_H_AU09_r
sd_M_AU26_r
min_H_AU06_r
md_H_AU07_r
md_M_AU26_r
mean_H_AU17_r
md_H_AU17_r
md_M_AU17_r
mean_M_AU17_r
min_M_AU45_r
sd_M_AU17_r
−0.5
0.0
0.5
1.0
Feature weight (CV2)


HEADsync
kurtosis_H_pose_Rysync
skew_H_pose_Rysync
mean_M_pose_Rysync
max_M_pose_Rzsync
md_H_pose_Rxsync
sd_M_pose_Rxsync
mean_H_pose_Rxsync
mean_M_pose_Rxsync
sd_M_headsync
max_M_pose_Rysync
sd_H_pose_Rxsync
max_H_headsync
sd_M_pose_Rysync
sd_H_headsync
min_H_headsync
min_M_headsync
−1.0
−0.5
0.0
0.5
Feature weight (CV2)

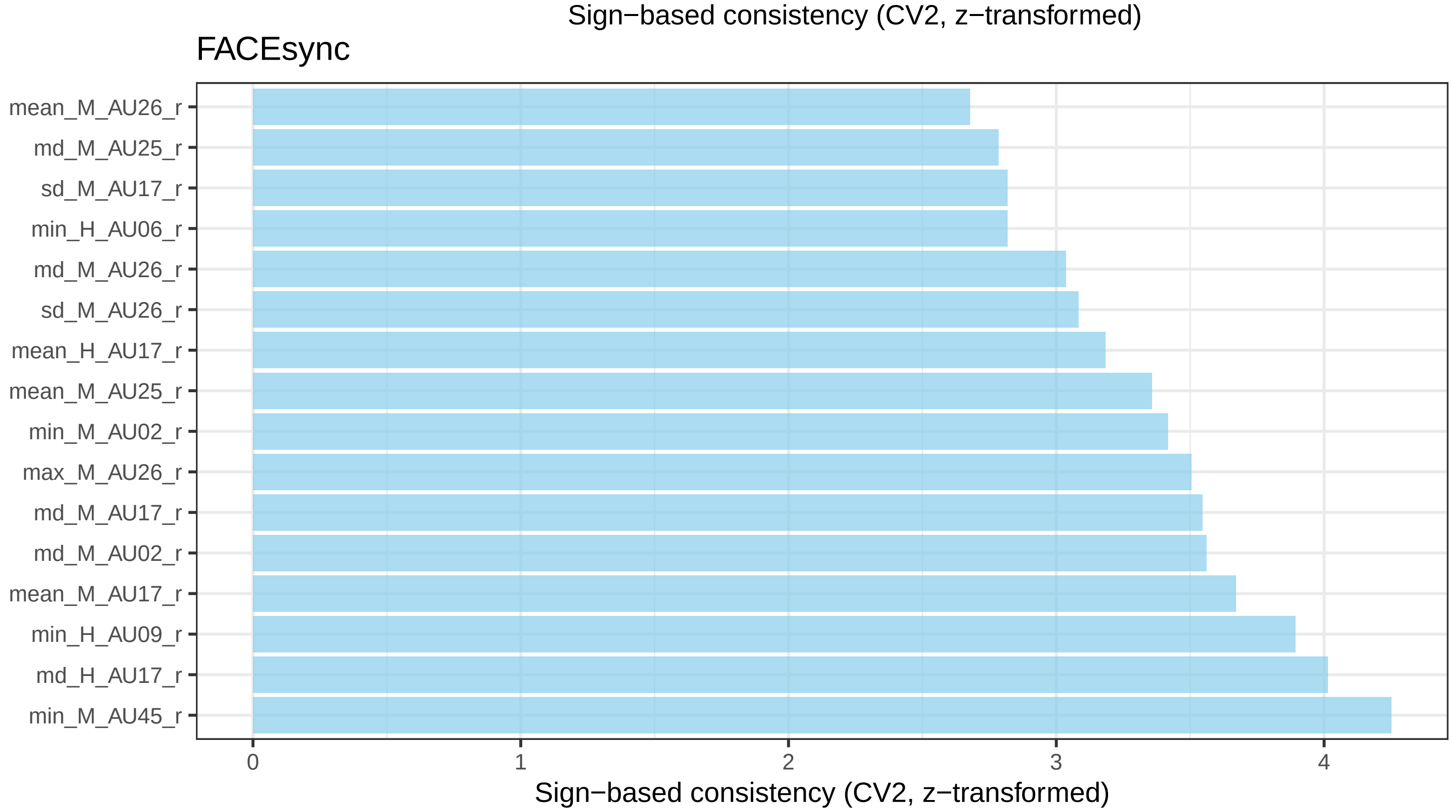

INTRAsync
md_M_intra
kurtosis_H_intra
skew_H_intra
min_H_intra
kurtosis_M_intra
skew_M_intra
min_M_intra
max_M_intra
sd_H_intra
max_H_intra
md_H_intra
mean_H_intra
sd_M_intra
mean_M_intra
−0.6
−0.3
0.0
0.3
0.6
Feature weight (CV2)

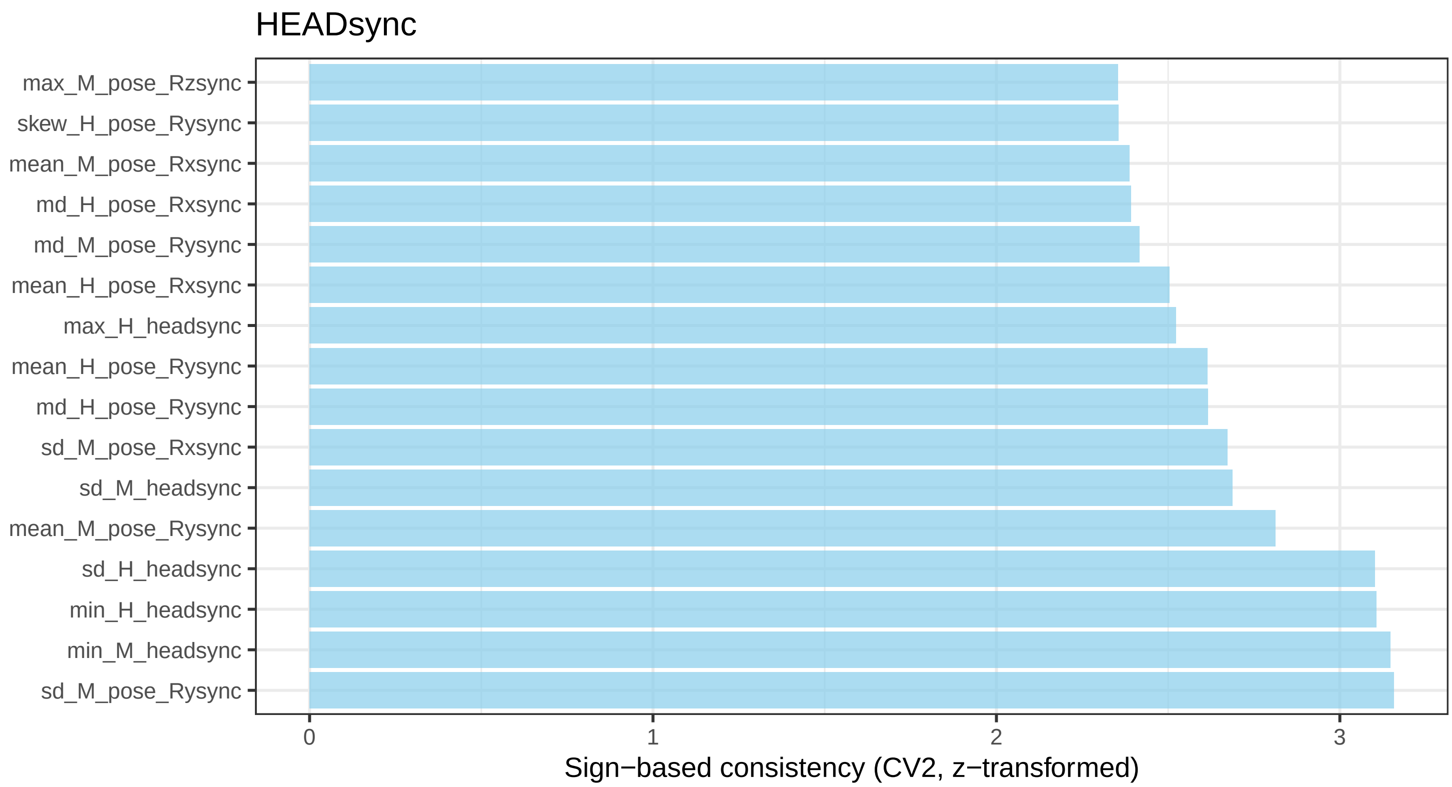

MovEx
M_body_total_movement
H_body_total_movement
H_head_total_movement
mean_intensity_M
mean_intensity_H
M_head_total_movement
−0.50
−0.25
0.00
0.25
Feature weight (CV2)
Speech
nsyll_M_speech
int_sync_H_speech
art_H_speech
dyad_str_M_speech
dyad_no_turns_H_speech
dyad_pit_sync_MEA_M_speech
pit_var_M_speech
dyad_pit_sync_MEA_H_speech
art_sync_M_speech
npause_H_speech
art_M_speech
dyad_str_H_speech
dyad_spr_H_speech
pit_sync_M_speech
dyad_int_sync_MEA_M_speech
dyad_spr_M_speech
−0.4
−0.2
0.0
0.2
Feature weight (CV2)

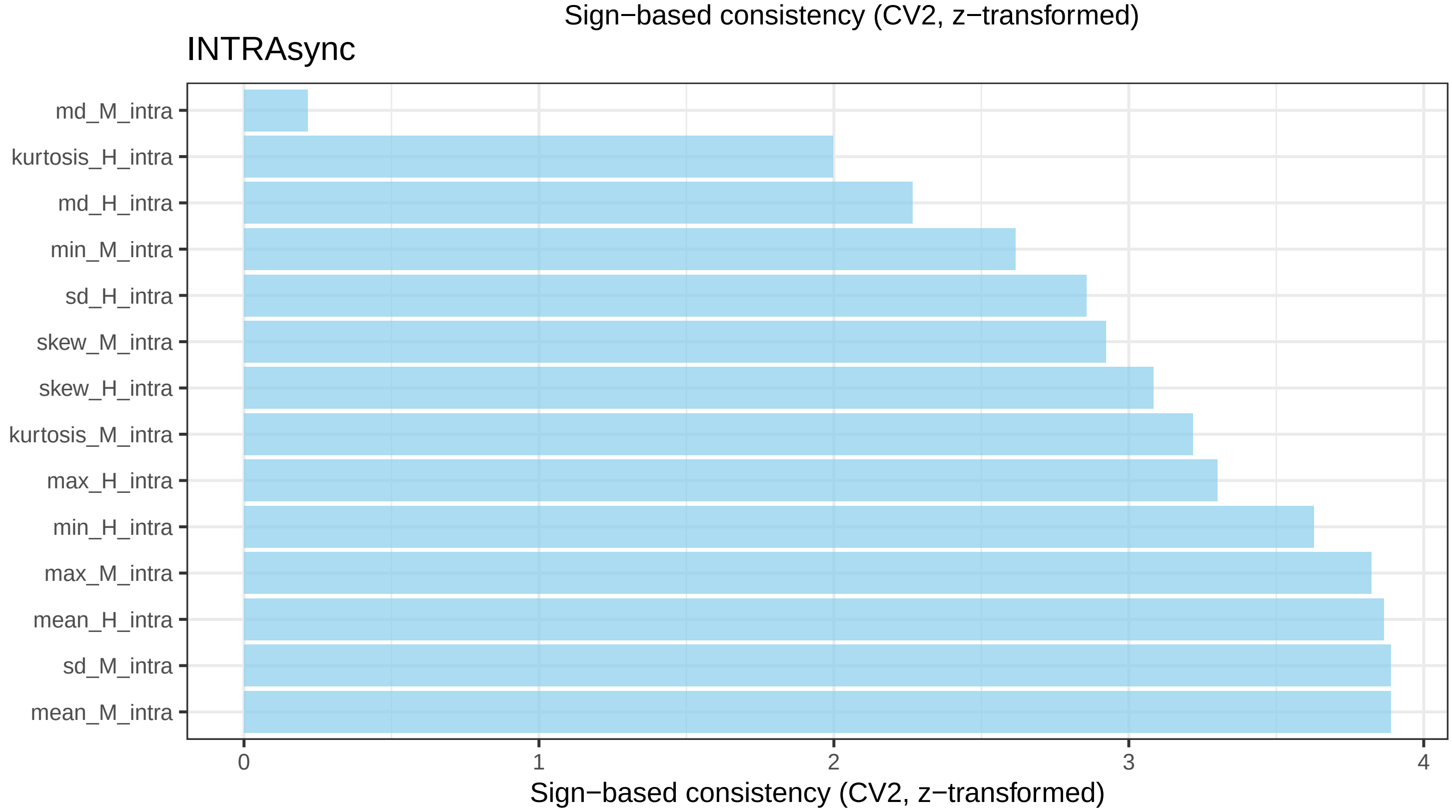

BODYsync
sd_M_bodysync
max_M_bodysync
mean_H_bodysync
mean_M_bodysync
kurtosis_H_bodysync
max_H_bodysync
min_M_bodysync
skew_M_bodysync
kurtosis_M_bodysync
sd_H_bodysync
skew_H_bodysync
md_H_bodysync
md_M_bodysync
min_H_bodysync
0
1
2
3
4
5
Sign−based consistency (CV2, z−transformed)

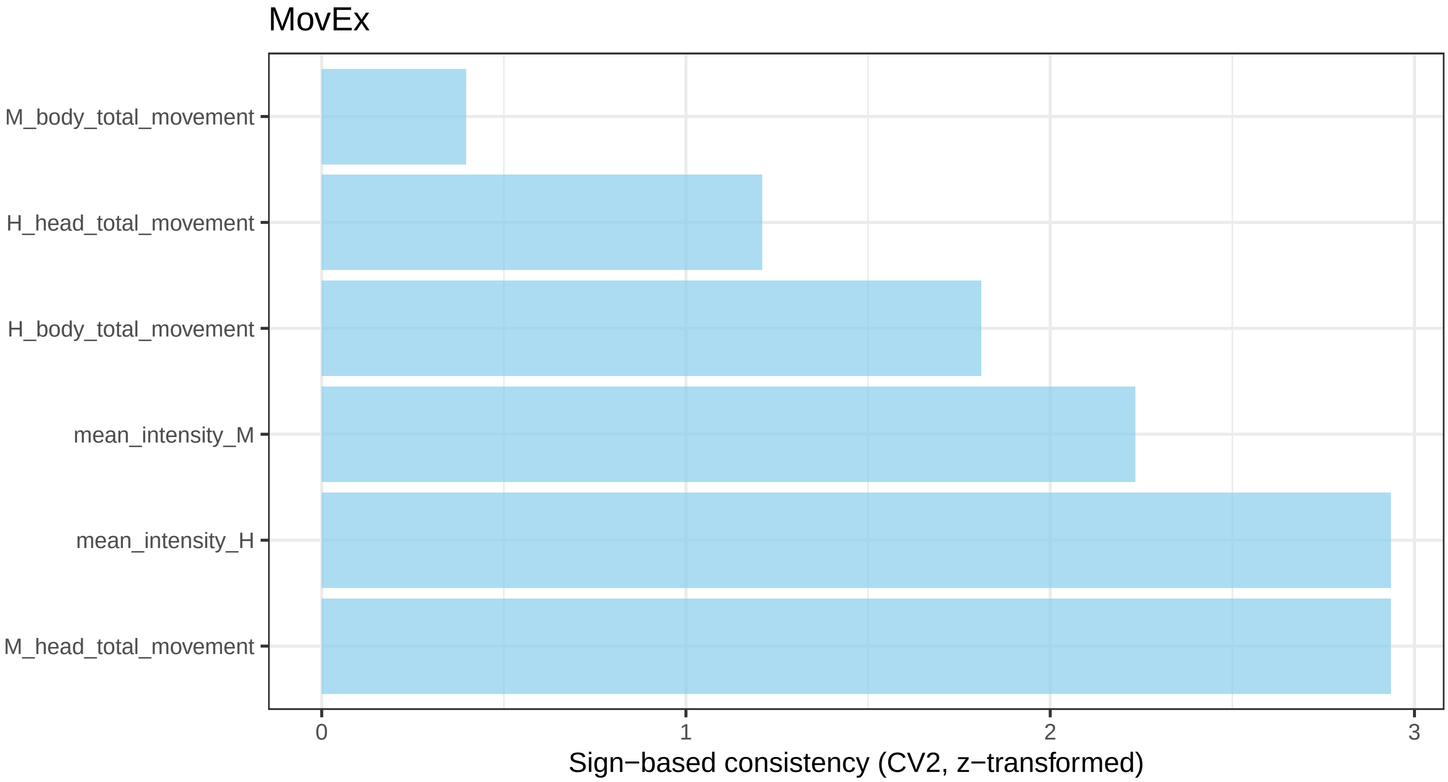

CROSSsync
md_H_LOF
skew_H_LOF
sd_M_LOF
md_M_LOF
mean_M_LOF
sd_H_LOF
min_M_LOF
sd_M_ROF
mean_M_ROF
min_H_ROF
kurtosis_H_LOF
kurtosis_H_ROF
mean_H_LOF
kurtosis_M_ROF
min_M_ROF
md_H_ROF
0
1
2
3
Sign−based consistency (CV2, z−transformed)

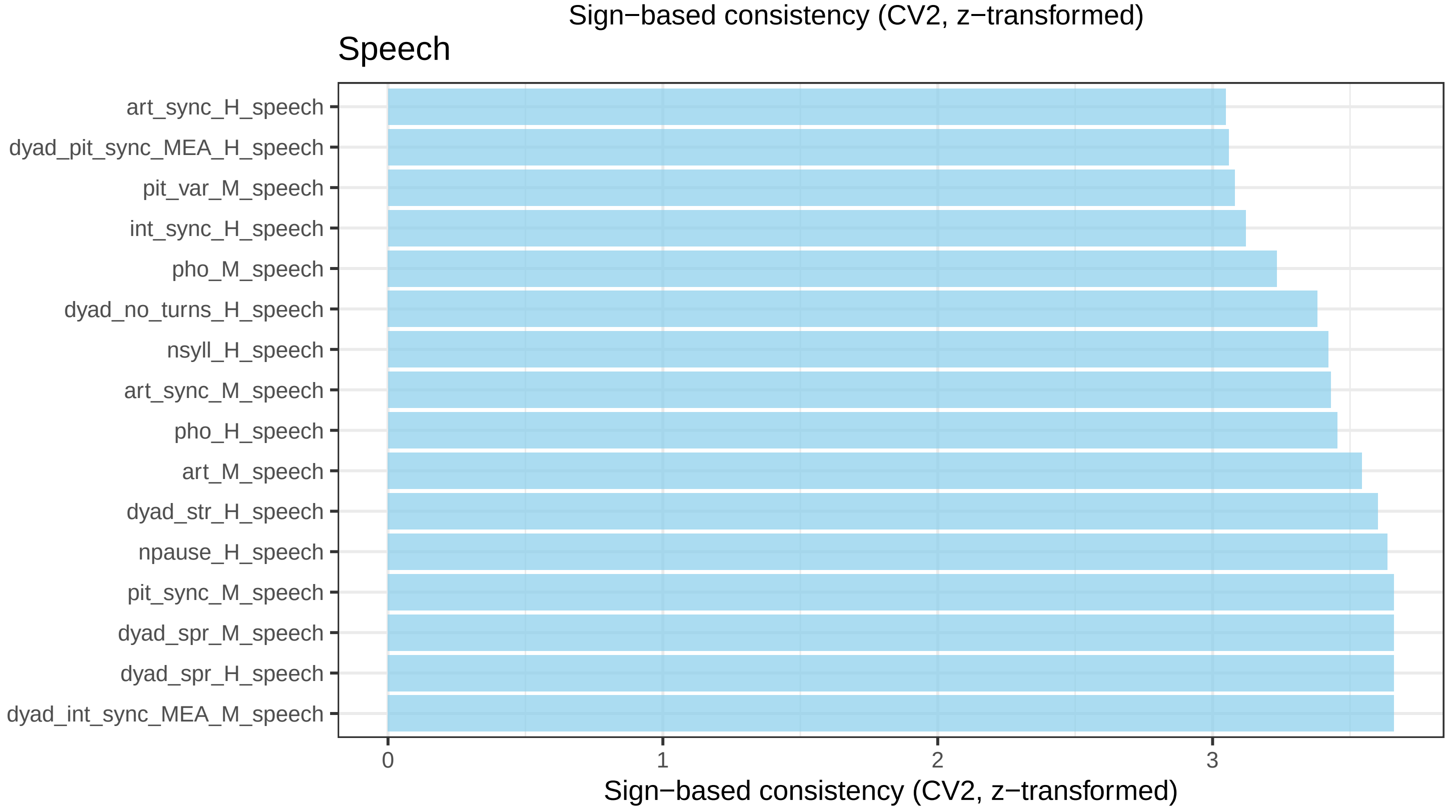

CROSSturn
self_max_H_AU01_r
other_max_H_AU01_r
self_mean_M_AU02_r
other_mean_M_AU02_r
self_sd_M_AU01_r
other_sd_M_AU01_r
self_mean_M_AU01_r
other_mean_M_AU01_r
self_sd_H_AU02_r
other_sd_H_AU02_r
self_mean_H_AU02_r
other_mean_H_AU02_r
self_mean_H_AU01_r
other_mean_H_AU01_r
self_sd_H_AU01_r
other_sd_H_AU01_r
0
1
2
3
4
Sign−based consistency (CV2, z−transformed)


FACEsync
mean_M_AU26_r
md_M_AU25_r
sd_M_AU17_r
min_H_AU06_r
md_M_AU26_r
sd_M_AU26_r
mean_H_AU17_r
mean_M_AU25_r
min_M_AU02_r
max_M_AU26_r
md_M_AU17_r
md_M_AU02_r
mean_M_AU17_r
min_H_AU09_r
md_H_AU17_r
min_M_AU45_r
0
1
2
3
4
Sign−based consistency (CV2, z−transformed)


HEADsync
max_M_pose_Rzsync
skew_H_pose_Rysync
mean_M_pose_Rxsync
md_H_pose_Rxsync
md_M_pose_Rysync
mean_H_pose_Rxsync
max_H_headsync
mean_H_pose_Rysync
md_H_pose_Rysync
sd_M_pose_Rxsync
sd_M_headsync
mean_M_pose_Rysync
sd_H_headsync
min_H_headsync
min_M_headsync
sd_M_pose_Rysync
0
1
2
3
Sign−based consistency (CV2, z−transformed)


INTRAsync
md_M_intra
kurtosis_H_intra
md_H_intra
min_M_intra
sd_H_intra
skew_M_intra
skew_H_intra
kurtosis_M_intra
max_H_intra
min_H_intra
max_M_intra
mean_H_intra
sd_M_intra
mean_M_intra
0
1
2
3
4
Sign−based consistency (CV2, z−transformed)





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